



THE REAL
FAIRY FOLK
• LOUISE JAMISON •



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THE REAL FAIRY FOLK



“‘I FEEL THE WIND,’
CRIED RUTH, WITH
BRIGHT EYES. ‘DEAR
VOICE, ARE YOU THE
WIND?’”

***THE
Real Fairy Folk***

BY
LOUISE JAMISON

ILLUSTRATED
BY
JAMES M. GLEESON



NEW YORK GARDEN CITY, N. Y.
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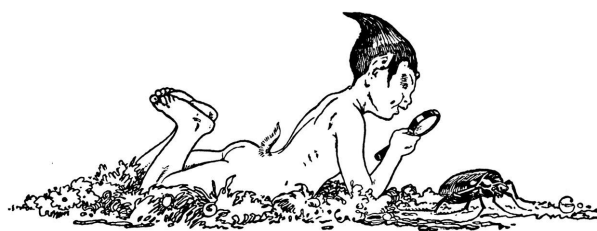
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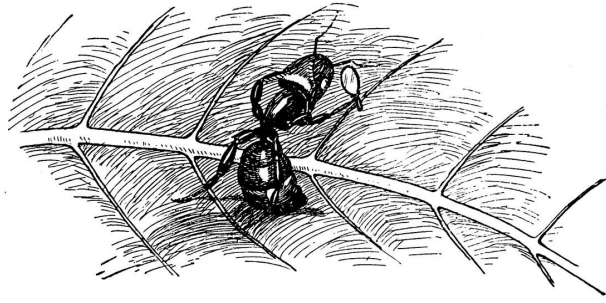
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To my Mother and Father this little book is lovingly dedicated



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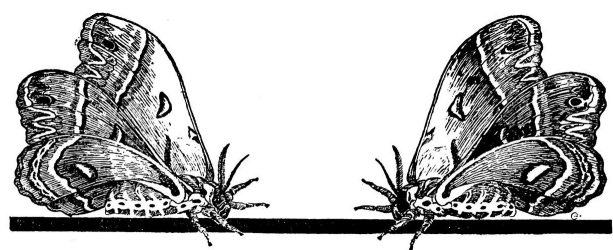
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THE REAL FAIRY FOLK



CHAPTER I

IN THE OLD WILLOW TREE

He prayeth best who loveth best
All things both great and small.

—Coleridge.

Ruth climbed to her favourite perch in the old willow tree, and settled Belinda in a crotch beside her.

“Now,” she said, drawing a long breath, “we will be cool and comfy.”

Certainly if there was a cool spot to be found on this hot August morning it was in the shade of this big willow.

“Her very own tree,” as Ruth always called it, for, since she could climb at all, she had loved to sit among its drooping branches and hear the leaves whispering together the wonderful things, which she knew they were telling each other, even though she could not understand them.

Then, too, she could look down into the brook, and watch the doings of the queer little people who made their home there.

These, like all the tiny folk of the outdoor world, were a source of never-failing interest and wonder.

In their company, Ruth was never lonely, even though she had neither brother nor sister, nor indeed any little boy or girl to play with.

Still it would be so much nicer if she could only talk to the bugs and things. There were such lots of questions she wanted to ask them.

How she did wish that the funny old tumble bugs would stop rolling their ball, and tell her all about it. They never did, though. They just kept at that ball as though it was the most important thing in the world.

Then she wanted to know what the bees whispered to the flowers as they buzzed above them, and whether the butterflies spoke to each other as they flew by in the sunshine.

There were the ants, too, always so busy, and in such a hurry. How fast they could run when any one upset their nest; and how funny they looked carrying those queer white bundles.

Mother had called these bundles the ants' babies, but Ruth thought them very odd babies, and she wondered if they had to be fed and bathed and put to sleep like human babies.

She wanted to know all about them, and about the spiders too, and their wonderful webs.

"Just think what a chance Miss Muffet had," she said to Belinda, when both were settled to her satisfaction in the willow-tree perch. "Only a very friendly spider would come up and sit down by you, and who knows the interesting things it could tell. The idea of being afraid of a spider anyhow! You might as well be afraid of that funny old toad in the garden, and I don't believe he could hurt you if he tried. I guess he doesn't do anything but sleep."

Ruth had been trying to talk to the toad that very morning. He had looked so solemn and so wise as he sat under the shade of a big stone in the damp corner of the garden, "but," as she said, "he wasn't any good at all," for he only looked at her, then drew a film over his eyes, and went on swallowing very hard.

"He can talk, though, I know," she said to Belinda. "They can all talk in their way. It sounds like noise to us, because we can't understand. Do hear them, Belinda? What are they saying?"

But of course Belinda could not answer. She never said more than "mama," in a very squeaky voice, and you had to squeeze her ever so hard to make her do that.

Ruth sighed softly, then, leaning forward with her elbow propped on her knee, and her chin resting in the palm of her hand, she listened to the flood of sound about her; the hum and buzz that came from garden and orchard, from field and meadow; thousands of tiny voices, rising and falling and rising again, as they told their fascinating life stories, from every leaf and twig and grass blade.

“They are talking just as fast as they can,” Ruth said again, “but I don’t know what they are saying. Oh! if I only did. Why don’t people learn their language instead of German and French and lots of other old things that aren’t any good? It would be ever so much nicer, and they could find out so many wonderful things, couldn’t they, Belinda?”

But, as usual, Belinda only stared at Ruth, and said nothing.



“SOMETIMES IT
SEEMS AS IF IT
MUST BE
FAIRYLAND ALL
AROUND, ONLY
I’M DEAF”

“Oh, dear,” said Ruth, “if you were only alive, and could tell me things, you’d be ever so much more interesting, but then maybe,” she added, thoughtfully, “I wouldn’t understand you any better than I do them. Maybe doll language is different too. It is all so puzzling. Sometimes it seems as if it must be Fairyland all around, only I’m deaf. I wonder if there’s a word that lets you in so you can know about things, like ‘Open Sesame’ in ‘The Forty Thieves.’ Oh, Belinda, do you think there is?” And Ruth clasped her hands together at the very thought. “But we can’t find it out,” she added, more soberly, “and so it wouldn’t be any use.”

“Watch and listen! Watch and listen!” said a voice so close to her ear that Ruth jumped, and nearly fell to the ground.

She looked about her expectantly, but no one was in sight, either in the tree or under it.

“It is very queer,” she said. “You can’t talk, Belinda, and I don’t see a single person anywhere.”

“It is not so queer as you think,” the voice replied, as close to her ear as before. “You cannot see me, but you can feel me.”

A passing breeze had touched her cheek and was softly ruffling her hair.

“I feel the wind,” cried Ruth, with bright eyes. “Dear voice, are you the Wind? Why have you never talked to me before? If you only knew how I have wanted some one to talk to me, and tell me things! People don’t seem to like to answer questions. They haven’t time or something. But you must know such a lot. The wind goes everywhere.”

“Yes, I am a great traveller, but, child, the marvellous things are not all far off. There is a wonderland right here at home, if one has the eyes to see, the ears to hear, and the heart to feel and understand.”

Ruth clapped her hands, and her eyes danced.

“I knew it! I knew it!” she cried eagerly. “I told Belinda it was Fairyland all around us; but, dear Wind,” she added, while a little cloud filled her eyes, “I do see and hear lots of things, but I *can’t* understand, and I *do* want to know all the whys and becauses. Won’t you please, *please* tell me?”

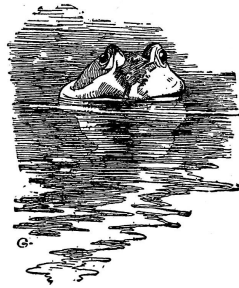
“I may not do that, child,” was the answer, “for each thing speaks in its own language, and will tell its own story to those who seek truly and earnestly. You are a thoughtful child, and for that reason it will be given to you to know those things which you most desire to learn. Only remember, ‘Watch and be patient,’ and never forget the password ‘Brotherhood,’ for even the lowest creature has some rights to be respected.”

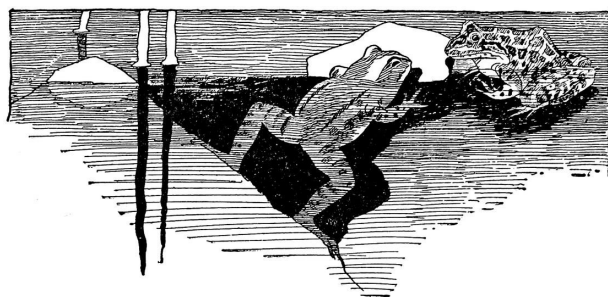
The breeze passed on, softly singing through the willow branches, but Ruth sat without moving, her eyes wide with eager wonder.

“I didn’t dream it,” she said at last in an awed little whisper. “It was as real as anything could be that you couldn’t see. I suppose ‘brotherhood’ means not to be unkind or cruel to things. Oh, Belinda, just think of it: hearing what they say, the bees and the butterflies and the dear little crickets and funny old grasshoppers,” and she snatched Belinda to her and hugged her tight. “It will be harder than ever to go into the house now, won’t it?” she finished soberly. Then she sat for a few minutes thinking, very quiet, but very happy.

“Kerchug—kerchug—kerchug,” called a voice from the brook, and Ruth started so suddenly she nearly dropped Belinda, and caught a branch just in time to keep herself from falling.

“Gracious,” she said, “how that scared me. I do believe it was that big green and brown frog. See him down there, Belinda? He is just showing his head and his funny eyes out of the water. Let’s get down close to him, and maybe he’ll come out all the way.”





CHAPTER II

TWO FUNNY GENTLEMEN AND WHAT THEY SAID

Nothing useless is or low.

—Tennyson.

“To be sure I’ll come out,” answered a croaky voice, as Ruth, holding Belinda tightly, drew close to the edge of the brook. “How’s that?” and with a splash a big green and brown frog landed on the stone at her feet.

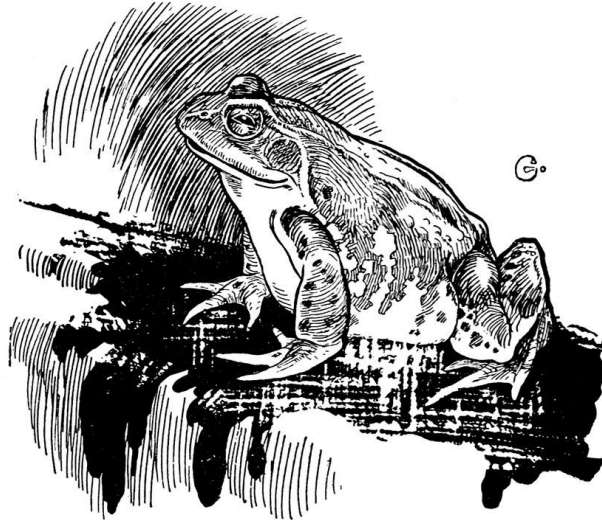
“Now,” he added, swelling out his white vest with an air of importance, “I am a frog, of course, but my family name is Rana. Please don’t forget it.”



“RUTH,
HOLDING
BELINDA
TIGHTLY, DREW
CLOSE TO THE
EDGE OF THE
BROOK”

“Family name?” said Ruth, sitting down on the edge of the stone. “I didn’t know frogs had family names.”

“There’s a great deal you don’t know,” said Mr. Rana, in his decided way.



“‘HOW’S THAT?’ AND WITH A SPLASH A
BIG GREEN AND BROWN FROG LANDED
ON THE STONE AT HER FEET”

“Maybe there is,” agreed Ruth, “but it isn’t very polite to tell me so.” Then, with a sudden thought, she added quickly, “Why, you are really talking.”

“Of course, I’m talking. Do you suppose it’s the first time?”



“I AM A FROG,
OF COURSE, BUT
MY FAMILY
NAME IS RANA”

“He’s dreadfully snappy,” Ruth whispered to Belinda.

“It isn’t my fault that people can’t understand,” finished Mr. Rana, swallowing very fast.

“I wanted to understand,” declared Ruth meekly. “I was sure you could tell me such a lot of interesting things, and that nice fat toad in the garden too. He is so——”

“You’d better talk to the fat toad, then,” said Mr. Rana, looking very cross.

“Oh, dear,” sighed Ruth, “I didn’t mean I’d *rather* talk to him. I do want you to tell me things. All about yourself, please.”

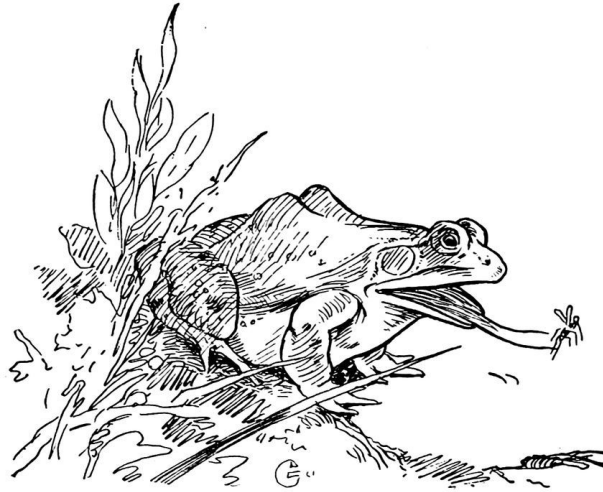
“Now you are showing your good sense,” said Mr. Rana, as Ruth settled herself with a ready-to-listen air. “Nothing can be more interesting than my story; but excuse me one second. I see Mrs. Mosquito. This morning I ate her husband, and now——”

His sentence was not finished, but Mrs. Mosquito was; and Mr. Rana folded his hands across his fat stomach and looked at Ruth, while a big smile played about his broad mouth.



“THAT NICE FAT TOAD IN THE GARDEN”

“She’s gone,” said Ruth, in a slightly awed tone, “and I know you’ve swallowed her, but I wish you would tell me how you did it. I didn’t see you move.”



“I DIDN’T MOVE, BUT MY TONGUE DID”

“I didn’t move, but my tongue *did*, and it went so quick you couldn’t see it. When you eat, you bring things to your tongue, but when I eat, I send my tongue to my dinner. It’s a simpler way, I think. My tongue is rather wonderful too. It is fastened to my mouth in front, and rolled back; besides, it has a sort of glue on the end that catches whatever there is to catch. The number of pests I eat in a day would astonish you. Slugs, grubs, snails, mosquitoes, and—well, what’s the matter? You don’t like such things, I suppose. Tastes differ, you see. Now, to tell my story. What do you think I looked like when I was first hatched?”

“A tadpole, of course,” answered Ruth. “I’ve seen lots of tadpoles. They are funny, wiggly things.”

“They *are* lively fellows,” agreed Mr. Rana, swallowing several times, while Ruth silently watched the sides of his neck puff out.

“Please tell me why you swallow so much,” she asked at last. “You are not eating, are you?”

Mr. Rana smiled, and this time the smile went all around his mouth.

“I swallow to breathe,” he answered. “I can’t swallow air while my mouth is open, and so I stop talking and shut it. Every time I swallow, the air sac on the side of my neck fills out. That’s why my voice has such a lovely croak. My poor wife hasn’t any air sac, so her voice is never croaky.”

“But in the water——” began Ruth.

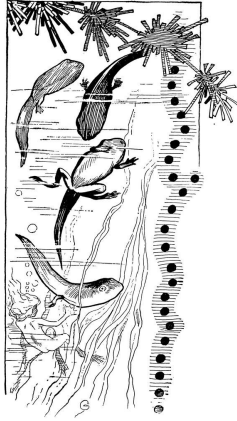
“In the water,” answered Mr. Rana, “I take in air through my skin. It is very porous. My skin I mean. It is really a pleasure to tell you things. Now to get back to the beginning, being a tadpole, or, I should say, an egg. Looking at me now, could you imagine that I was once a tiny egg? It’s a fact, though. My mother laid her eggs near some water rushes, and, as I said, these eggs were but tiny specks, black specks enclosed in a gluey case, which the water made swell, until it looked like a mass of jelly. I came from one of those specks, and I tell you I was a lively fellow when I was first hatched. Some people say tadpoles are all head and tail, but there were other parts to me—places for legs, and I know I had two eyes and a mouth. Of course I made the most of life. A whole pond to circle in seemed a mighty big world to me, and I was soon swimming about with a lot of other tads, slapping tails, and having all kinds of fun. Indeed, we were always lively, especially when we were trying to get away from those who wanted us for dinner. There were lots of them too.”

“Ugh!” said Ruth, screwing up her face.

This displeased Mr. Rana.

“A tadpole is very delicate eating,” he said. “You have never tasted one, so you cannot judge; but let that pass. *I* was not eaten, as you can see for yourself.”

“I am glad you were not,” said Ruth as Mr. Rana stopped to swallow some air, “because then I shouldn’t have known you.”



“I WAS SOON
SWIMMING
ABOUT WITH A
LOT OF OTHER
TADS, SLAPPING
TAILS, AND
HAVING ALL
KINDS OF FUN”

“Well, that’s a fact. Now let me see what comes next. Oh, yes—my legs. Legs, you must know, are very important affairs to a tadpole, because when he gets them he isn’t a tadpole any more; so you may be sure I was happy when I saw mine beginning to grow. At the same time, my tail became shorter and shorter, until at last I had none at all. I was really and truly a frog. After this I was not obliged to stay in the water all the time. I had lungs and could breathe air.”



“A LOUD SPLASH AND MR. RANA’S LONG
LEGS DISAPPEARED IN THE BROOK”

“But you do go in sometimes,” said Ruth. “I’ve seen you.”

“Of course I do,” agreed Mr. Rana. “I must keep my skin wet, and that reminds me it’s pretty dry now, so I will have to leave you. Good-bye for the present.” And before Ruth could say a word there was a loud splash and Mr. Rana’s long legs disappeared in the brook.



“I’M RIGHT
OVER HERE IN
THE SHADE”

“Oh, dear, he’s gone!” sighed Ruth.

“Yes, and good riddance,” croaked a voice that was not Mr. Rana’s.

Ruth looked around quickly.

“It’s nice having things talk to you,” she said, “but it keeps you jumping.”

“Use your eyes, and you wouldn’t have to jump,” went on the same voice. “I’m right over here in the shade. My blood’s cold, and I can’t stand the hot sun.”

It was her friend the garden toad. Ruth could see him plainly now. He looked more puffy than ever, as he sat under the bushes, swelling his leathery throat with importance. “If my cousin can talk to you I guess I can too,” he added. “I’m Mr. Bufo, and I’m quite as interesting as he is.”

Ruth was only too willing to agree to this, though, as she whispered to Belinda, she thought frogs and toads had very good opinions of themselves.

“I have a wife,” croaked Mr. Bufo when Ruth had sat herself on the ground close to him, “a worrying wife. Do you know it’s a bad thing to have a worrying wife?”

Ruth didn’t know, but she nodded her head in agreement.

“A bad thing,” repeated Mr. Bufo. “In the Spring, after Mrs. Bufo had laid her eggs, she gave me no peace. Of course, like all toads, she laid them in the water, but, instead of being reasonable about it, she was always asking me how she was to know them from the eggs Mrs. Rana and Mrs. Urodillo had laid. Theirs were in the water too.”

“Please, who is Mrs. Urodillo?” asked Ruth. “I know Mrs. Rana is a frog.”

“Mrs. Urodillo is a water salamander,” answered Mr. Bufo, not over pleased at being interrupted. “Now where was I? Oh, yes. Mrs. Bufo was afraid she wouldn’t know her own eggs. Well, I tried to argue with her.”

“‘Didn’t you lay yours in double strings?’ I asked, ‘and didn’t you with motherly care enclose them in thin but strong tubes?’ Of course she couldn’t deny it. ‘But I won’t know my own tadpoles,’ she kept insisting.”

“No wonder she was worried,” said Ruth. “Any one would want to know their own babies.”

“Mothers in our family never do,” declared Mr. Bufo. “They lay their eggs, and that’s the end of it. Mrs. Bufo knew that as well as I did. She only wanted something to worry about. All tadpoles are pretty much alike to begin with, but they don’t end alike. Toad egg tads always grow into toads; frog egg tads become frogs, and salamander egg tads will be salamanders and nothing else.”

All the while he talked Mr. Bufo had stopped every little while to swallow, not only air, but whatever in the way of insects came within his reach. So of course Ruth saw his tongue.

“Your tongue is just like Mr. Rana’s,” she said, after watching it for a few seconds.

“Our tongues may be alike,” agreed Mr. Bufo, “but there’s a vast difference in our legs. His are too long for any use, and his skin is so horribly smooth it gives me the shivers just to look at it. Of course I know I am not handsome, and that reminds me of some lines that have been written about me. Want to hear them?”

Then without waiting for an answer he swallowed some air and began:

“I’m a clumsy, awkward toad,
And I hop along the road;
'Tis the only way we toads can well meander;
While in yonder marshy bog
Leaps my relative the frog,
Very near my aunt, the water Salamander.

“And if you should ever stray
Near a slimy pool some day,
And along its grassy margin chance to loiter.
Do not pass it idly by,
For it is the spot where I
Was born a lively tadpole in the water.

“I’m a homely, harmless thing;
I catch insects on the wing,
And in this I serve you all; it is my duty.
And now tell me which is best,
To be useless and well dressed,
Or useful, even though I am no beauty?”

Mr. Bufo had scarcely finished, when his mate hopped out from some nearby bushes.

"I'd be ashamed," she said, in a very puffy voice, "to sit there repeating that lovely poetry, with such shabby clothes as yours are. How many more times must I tell you to change them?"

"It doesn't matter about his clothes," said Ruth. "I think it is so lovely to hear him talk."

"You haven't heard him as often as I have," puffed Mrs. Bufo, hopping almost into Ruth's lap. "Besides, his clothes are a disgrace. They are not only faded and dull, but they are actually beginning to split up the back."

"Are they?" croaked Mr. Bufo meekly.

Then he drew a film over his eyes and pretended to be asleep.

"Now look here," said Mrs. Bufo, "you can't deceive me. That is only your third eyelid. You are not asleep. Now do get off those old clothes."

"Well, if I must, I must," croaked Mr. Bufo, hopping away.

"There, I've made him do it at last," puffed Mrs. Bufo, swallowing a passing fly. "It's a hard job, and I don't blame him for getting out of it as long as possible. He has to twist and turn, and use first one leg and then another, until he is quite free from his old suit, and then, tired as he is, he must eat it."

"Eat it?" repeated Ruth, screwing up her face.

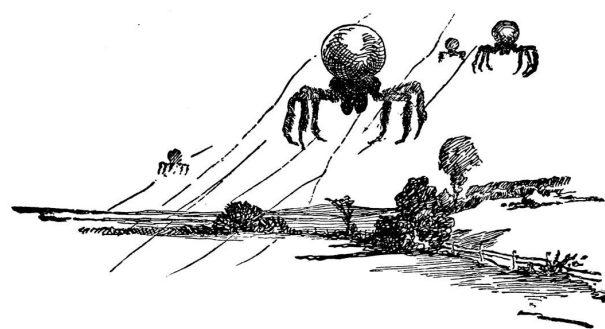
"Yes, eat it, and not a tooth to chew with either. I can't see why we haven't teeth like those horrid frogs, though, to tell the truth, theirs are no good for chewing. They only have them in their upper jaws, and they point backward, too, like fish teeth. I can't see that they help much in chewing, but they do help to hold what the frog wishes to swallow, and, after all, we toads and frogs are swallowers rather than chewers."

As she spoke, several flies went to prove her words.

"Yes," she added with a big puff, which Ruth took for a sigh, "we have our troubles and worries from early Spring, when we leave our holes, where we sleep all Winter, to the time when frost drives us into our holes again, and no one seems to think about the work we do. The garden couldn't have a better friend, for the bugs and harmful insects we eat can't be counted. Well, there's no use talking this way. I must go to Mr. Bufo. He'll need some cheering up, I'm sure. One good thing, he won't have to make his new suit. He'll find it all ready under his old one."

“Well, she does think of him, anyhow,” thought Ruth as Mrs. Bufo hopped away. “I hope she will talk to me again some day.”





CHAPTER III

RUTH AND THE WONDERFUL SPINNERS

She throws a web upon the air and soon
'Tis caught and lifted by the willing breezes,
Then, freed from trouble in her light balloon,
Our spinner travels wheresoe'er she pleases.
—*Edith M. Thomas.*

Ruth was in the garden counting colours among the hollyhocks when a little breeze hurried by.

“Come,” it said, kissing her cheek, “and hurry; things are going to happen.”

“It is my dear Wind,” cried Ruth, her eyes growing big with expectation, and, stopping just long enough to snatch up Belinda, who of course would wish to go, too, she followed where the little breeze led.

This was to a lovely spot on the edge of the wood, and one of the first things she saw was a big round spider’s web on the branches of a tall bush.

“Oh,” she said, going up closer, “who would ever think a spider could make anything like that?”

“Indeed,” said a voice which made her give a little jump, “who else but a spider could spin a web, I’d like to know? You haven’t any brains, I’m thinking.”

“Oh, please excuse me,” said Ruth. “I didn’t know you were there.”

“That’s because you don’t use your eyes properly,” was the answer of the large, handsome black and gold spider hanging head down from the centre of the big web.

Her eight long, slender legs were outstretched and rested by their tips on the bases of the taut radii, and her eight eyes were staring at Ruth.

“I saw you as soon as you came,” she said.

“I suppose you will stay to the meeting. I’m to be chair-spider.”

“Chair-spider?” repeated Ruth, slightly confused by those eight bright eyes. “And please, what meeting?”

“Why, our meeting, of course. Mrs. Cobweb Weaver says men always have a chairman at their meetings, so why shouldn’t spiders have a chair-spider, I’d like to know?”

“I suppose they should,” agreed Ruth.

“Of course we should. Considering you are a human creature, with only two eyes, two legs, and no spinnerets, you really show a great deal of sense. Now sit down on the crotch of that little tree, then you will be near me and can hear all I say. What’s that thing you are carrying?”

“Why, it’s Belinda, my doll,” explained Ruth. “I tell her everything. I think she will like your—your—meeting.”

“Well, I don’t care whether she does or not,” said Madame Spider. “Now our friends are arriving, and as you can see, with even two eyes, they are all shapes and sizes. Long legged, short legged, plump, thin, grave and gay. All colours too—quite enough to satisfy any taste, I should say.”

Ruth looked about her in wide-eyed astonishment.

“I never knew there were so many kinds of spiders,” she said at last, “or that they had such lovely colours. I thought spiders were mostly grayish or brownish.”

“That is because you haven’t used your eyes, as I said before; but you are only like others of your kind. Such ignorance! Because some spiders are dull and colourless, most people imagine that all are so. I suppose they think, if they stop to think at all, that all kinds of webs are spun by the same kind of spider, and that all spiders spin webs.”

“Don’t they?” asked Ruth, with some hesitation, for Mrs. Spider’s indignation made her look quite fierce.

“They do *not*,” was the decided answer. “All spiders are spinners, but not all are web makers.”

Ruth looked puzzled.

“You see,” explained Mrs. Spider, “it all depends upon the way they catch their prey. Spider habits are as different as their looks. Some like the sun, others prefer the shade. Some live in the forest, and others with the house people. Many make their home in the bark of trees, and under stones.”

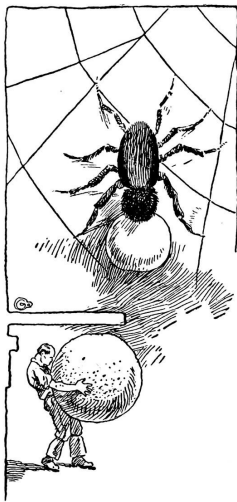
“I’ve seen that kind,” interrupted Ruth, eagerly, “and when you lift up the stone they run awfully fast. Sometimes they have a funny little gray bundle, just as the ants carry their babies. Maybe it’s their babies too. Is it?”

“Well, they will be babies if nothing happens. Those gray bundles are cocoons full of eggs. The mother spins the cocoon of silk from her own body.”

“Oh, now, I understand. They are spinners, but they don’t have any web. Isn’t that it?”

“Exactly. They do not need a web. They spring on their prey when the prey isn’t looking. We call them hunters, also runners.”

“Well, they *can* run,” said Ruth.



“THE MOTHER
SPINS THE
COCOON OF
SILK FROM HER
OWN BODY”

“The flower spiders are not web spinners either,” went on Madame Spider, who seemed to like nothing better than to talk. “They live among flowers, and eat the visiting insects. You can see some of them over there. Talk about colours! They are gay enough, just like flowers themselves. Perhaps you can guess why.”

Ruth thought a few minutes.

“Well,” she said, “if they were the same colour as the flower they couldn’t be seen so easily. I saw something walk out of an ear of corn once, and it looked like a kernel of corn on eight legs. It was awful funny. Was that a spider?”

“Very likely. We are wonderful enough for anything. I suppose you have never heard of the trapdoor spider and his silk-lined burrow, with its little hinged door, nor of the spider who lives under the water, in a tiny silken house, which she spins herself, and fills with air carried down, bubble by bubble, from the surface. Don’t look as though you didn’t believe me. It isn’t polite. I am telling you the truth. Very likely you’ll doubt me when I say that we

sail in balloons, of our own making, and cross streams of water on bridges, which we can fashion as we need them—that is, we orb weavers do, for, after all, we stand at the head of the spider clan. Did you know I was an orb weaver?”

“I—I—haven’t thought about it,” said Ruth, slowly, for the question had come very suddenly, “but I’d like you to go on telling me things. Do you always hang with your head down? I should think it would make you dizzy.”

“Dizzy? Whoever heard of such a thing? Of course I keep my head down, and my toes on my telegraph lines. Then I can feel the least tremble in any one of them, and I’m pretty quick to run where I know my dinner is waiting. Sometimes I don’t hurry quite so fast. That is when the line trembles in a way which lets me know that something big has been caught. Indeed, there are times when I bite the threads around what might have been my dinner, and let it go; for it is wiser to lose a meal than run the chance of being a meal.” And Mrs. Orb Weaver winked, not with one eye only, but with all eight. “Now it is time to talk to the company,” she added, “as I am chair-spider.”

She said the last words in a loud voice, intended for all to hear; then she looked around to see if any one objected.

“They had better not,” she said to Ruth, and in a louder voice, added: “My friends, we are not appreciated. Men talk about the wonderful bees, the wonderful wasps, the wonderful ants, but few of them say anything about the wonderful spiders. Now we are wonderful, too, and we are honest, and we are industrious. We eat flies and lots of other pests, and we do not hurt orchards, or steal into pantries, or chew up clothes. Indeed, we do man no harm at all. But is he grateful? Tell me that. I’ll tell you he isn’t. Ask Mrs. Cobweb Weaver if there isn’t always some broom sweeping down the nice web she makes. I wonder she doesn’t hate a broom. No, my friends, man is not grateful. Even those who call themselves our friends are ready to pop us into bottles, or boxes, whenever they get a chance. They give us what they call a painless death in the cause of science. Now we would rather live in our own cause. At least I would.”

Mrs. Orb Weaver had become so excited that her whole web was shaking violently.

Ruth was excited, too.

“It’s rather horrid to do that way,” she said, “but maybe people don’t know about you. I didn’t until to-day. The wonderful things I mean, and I want to know lots more. How your web is made and—and—everything. Please tell me.”

“Why, certainly,” answered Mrs. Orb Weaver readily. “To begin with, my web is made of silk.”

“Who didn’t know that?” snapped a running spider.

“I didn’t,” answered Ruth.

“You! And who are you, pray?”

“Be quiet,” commanded Mrs. Orb Weaver. “She is my guest, and anything she wishes to know I shall be happy to tell her. Now, to get on, our webs are made of silk, and the silk comes from our own bodies, through little tubes called spinnerets. It is soft at first, but gets harder when it reaches the air, just like caterpillar silk. We guide each thread with our hind feet, making heavier strands by twisting a number of fine ones together. Of course, we spin the foundation lines first. They are the ones which fix the web to the bush. Then the ray lines, those like the spokes in a wheel. They are all heavy strands, and only after they are finished do we spin the real snare, the lines which run around. They are very fine, and are covered with a sort of glue, for they have to catch and hold the flies and other insects that come on the web. We orb weavers are the only ones who have this glue. No other spiders use it. They trust to the meshes of the web to entangle their prey.”

“But why don’t the sticky parts catch you too?” asked Ruth, who had been listening with eager attention. “I’ve seen you run all over your web and——”

“We never get caught. Of course not,” finished Mrs. Orb Weaver. “And why? That’s a question. The wise men don’t know, and if we do, we are not telling. Now I am getting hungry, so I think I will tell a little story, then we will adjourn. I am sorry there isn’t time for Mrs. Funnel Weaver to speak.”

“But there is,” declared a large brown spider, whose body looked as though it were set on a framework of legs. “I mean to speak too—if only to point out all those webs in the grass.”

“Oh, I’ve often seen webs like that,” said Ruth. “They are lovely with dew on them. But why do you call yourself a funnel weaver?”

“I don’t!” she snapped. “The men, who think they know everything, gave me that name, because at one side of my web is a funnel-shaped tube. It is our way to escape our enemies. We run through it into the grass when something too big for us to manage gets into our web.”

“I generally make my web in houses,” said a small, slender-legged, light-coloured spider.

She spoke in a hurry, as though she was afraid some one might stop her before she finished. “I have cousins who like fields and fences and outbuildings, but our webs are all the same pattern. Not so regular as yours, Mrs. Orb Weaver, but very fine and delicate.”

“Oh, everybody knows you, Mrs. Cobweb Weaver,” said a voice from a nearby twig. “Now if you are speaking of legs——”

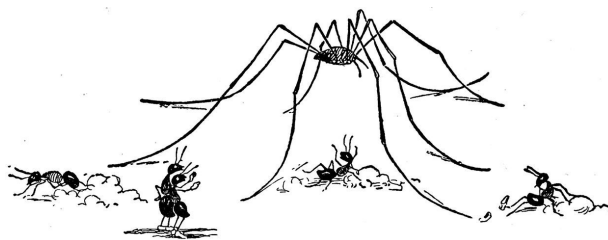
“We are not,” answered Mrs. Orb Weaver, “and I should like to know how you came here.”

“On my legs of course. Don’t you think they are long enough? And though I can neither spin nor weave, I am your relation, and I have as much right to be here as you have. I——”

“Why, it’s Daddy Long Legs,” interrupted Ruth, with a friendly smile of recognition. “I like daddies.”

“Well, I am not saying anything about my legs,” remarked a fat little spider, as Daddy tried to bow to Ruth, “though I have eight of them. I usually travel in a balloon, which I make myself. Oh, I tell you, it is fine to go

“Sailing mid the golden air
In skiffs of yielding gossamer.”



““WHY, IT’S DADDY LONG LEGS””

“Poetry,” said a handsome spider, wheeling back and forth on a silken bridge swung between two bushes. “I could have learned some too, but I didn’t know it was allowed. Of course I can build bridges. Who is asking that idiotic question? You?” And eight glaring eyes were fixed upon Ruth. “Maybe you don’t know that spiders were the first bridge builders and when men suspend their great bridges to-day they follow our ideas and ways, without giving us the least credit; but that’s the way with men.”

“Well, we can’t expect to regulate men,” answered Mrs. Orb Weaver, “and, besides, it’s time to tell my story, and then you will know why we get our name, and why we are such wonderful spinners. Now listen, all of you:

“Once upon a time——”

Ruth chuckled contentedly. All nice stories began, “Once upon a time.” “Please go on,” she whispered eagerly.

“Then don’t interrupt me,” said Mrs. Orb Weaver, and she began again:

“Once upon a time, ever so long ago, there lived in a beautiful land called Greece a maiden named Arachne. Arachne was not only fair to look upon, but she could also spin and weave in a fashion so wondrously fine that all who saw her work said that the great Athena herself must have been her teacher. Now this surely was praise enough, but Arachne was vain. ‘Nay,’ she said, ‘no one has taught me, and gladly will I weave with the great goddess herself, and thus prove the skill to be all my own.’ Her words only shocked all who heard them, but Arachne cared not, and again repeated her wish to try her skill with Athena.

“So it happened that as she sat spinning one day an old woman, leaning on a staff, stopped by her loom.

“‘Child,’ she said in a gentle voice, ‘a great gift is yours.’

“Arachne tossed her head, and answered scornfully:

“‘Well do I know it, yet Athena dares not try her skill with mine.’

“‘Dares not?’ repeated the old dame, in tones that should have made Arachne tremble. ‘Dares not, say you? Foolish maiden, be warned in time.’

“But Arachne was too proud to yield, and she still persisted, even though the old dame had dropped her mantle, and stood revealed as the great goddess herself.

“‘Be it so,’ said Athena, sternly, and both began to weave.

“For hours their shuttles flew in and out. Arachne’s work was wonderful, but for her theme she had chosen the weakness and the failure of the gods. Athena pictured forth their greatness. The sky was her loom, and from the rainbow she chose her colours, and when her work was finished and its splendours spanned the heavens, Arachne realized that she had failed.

“Ashamed and miserable, she sought to hang herself in the meshes of her web.

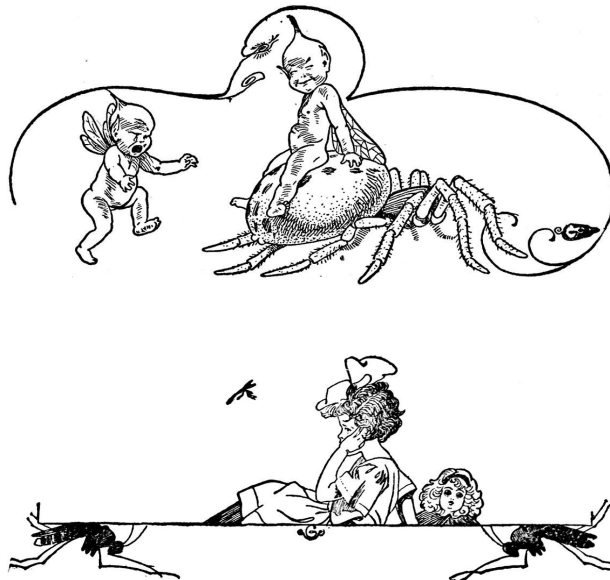
“‘Nay, rash maid,’ spoke Athena; ‘thou shalt not die, but live to be the mother of a great race, the most wonderful spinners on earth.’

“Even as Athena spoke, Arachne grew smaller and smaller, until not a maiden, but a spider, hung from that marvellous web.

“And now, my friends,” finished Mrs. Orb Weaver, “need I tell you that we are the wonderful race of which Athena spoke, and need *I* add that we have inherited Arachne’s marvellous skill, and are truly the most wonderful spinners on earth? Now I am hungry and the meeting is adjourned.”

“So am I,” added Daddy Long Legs, “not adjourned, but hungry, and, by the way, do you imagine any one believes that old story?”

He winked at Ruth, and then moved away as fast as his long legs would carry him.



CHAPTER IV

MRS. MOSQUITO AND HER KIN

“Thou art welcome to the town, but why come here
To bleed a fellow poet gaunt like thee?
Alas! the little blood I have is dear,
And thin will be the banquet drawn from me.”

—Bryant.

“**T**hat horrid mosquito,” said Ruth, waking with a start, and slapping her cheek.

“Aha! you didn’t get me that time,” answered a thin, high-pitched voice!

Ruth sat up. She had been asleep under the apple tree, but she was quite awake now.

“Where are you?” she asked, “and are you really talking?”

“I seem to be,” answered the mosquito, “though you tried to finish me just now. I bear no ill-will, though. I am quite used to being an outlaw. What is more, I don’t intend to be any better. I shall go on biting people as much as I please. I must have my meals as well as the rest of the world. People seem to forget that fact.”

“But just biting people——” began Ruth.

“It isn’t just biting,” put in the mosquito. “It really isn’t biting at all. I have a sharp little instrument to pierce the skin of the fellow I choose for my dinner, and the best kind of sucking pump to pump up his blood. That’s the way I get my meals. It is different with my mate. He is a harmless sort of fellow. He can’t even sing, and he likes such baby food as the nectar of flowers. Now tell me why I am different from other insect musicians.”

She fixed her big eyes on Ruth, who moved uneasily, and answered with not a little hesitation:

“I—I—really don’t know.”

“I’m a female. That’s why. In all the orders, so far as I know, the singers are males. Naturally I am proud of being an exception. Well, you didn’t know that. Do you know why I don’t care for science?”

“It is just like an examination,” thought Ruth, and again she answered.

“I don’t know.”

“Of course you don’t,” said Mrs. Mosquito. “Is there anything you *do* know? Well, I suppose I must tell you. I don’t care for science, because it interferes too much. Once upon a time men were our friends. We not only had nice juicy meals from them, but we had their rain barrels as nurseries for our children. Of course, what they said about us, when we came too near them, was not always complimentary, but a mosquito, attending strictly to business, doesn’t mind a little thing like that. But now come these fellows who know so much, or think they know so much. We carry malaria, they say, whatever that is, and the rain barrel must go, because it helps to breed mosquitoes. Not only that, these interfering fellows seem to spend their time thinking up ways to finish us. Well, I sting them every chance I get.”

“But alas! the rain barrel is going. I was hatched in one of the few to be found in these sad days. I was a lively baby, I can tell you. Young mosquitoes are called wrigglers and, true to my name, I wriggled for all I was worth. Now, when you know that my mother had laid something like three hundred eggs, and all had hatched into wrigglers as lively as myself, you can imagine the time there was in that old rain barrel.”

“But why,” asked Ruth “are you called wrigglers when you are young, and mosquitoes when you are grown up?”

“Why are you called baby when you are born, girl when you are half grown, and woman when you are quite grown?” replied Mrs. Mosquito, and Ruth said no more.

“Now,” went on Mrs. Mosquito, “I should like to tell you more about wrigglers, how they stand on their heads and breathe with their tails, and how they shed their skins when they become full-grown mosquitoes, but I haven’t time. The others are coming.”

“Others?” repeated Ruth. “What others?”

“The members of the Diptera order of course,” answered Mrs. Mosquito, with an important air. “You see, I found you sleeping under the tree and I

knew you wanted to learn about the things that are worth while, and as we are very worth while, I sent a friend to tell all the members of our order to meet in this spot.”

“Exactly what that young mosquito told me,” said Mrs. Hessian Fly, buzzing up excitedly.

She was a dusky-winged creature, scarcely more than an eighth of an inch long.

“What is the Diptera anyhow?”

“Why, you are one,” explained Mrs. Mosquito, with a superior smile. “It is quite a tax to know things for everybody,” she said to Ruth, “but you see I am around men so much I learn a great deal. I once attended a meeting of the men who think themselves wise. I wasn’t invited, you understand, but I went, and I attracted much attention too. Well, this is what I heard: The audience will please listen, it concerns you all:

“‘The members of the order Diptera have two gauzy wings and two thread-like organs with knobs at the end in the place where most other insects have a second pair of wings. Their mouth is framed for sucking, and sometimes for piercing. Only a few make cocoons. Their larvæ are called maggots, and they have no legs. Some are vegetable eaters, some carnivorous, and many are scavengers.’ They said all that about us, and maybe it’s true, but I tell you every man in that meeting felt my sting.”

“I don’t care what they say,” remarked Mrs. Hessian Fly. “To be talked about shows our importance, though I have never doubted mine. My family is a Revolutionary one, as my ancestors came over with the Hessians. Of course you have heard of them?”

“No, I am only interested in the people who live now,” answered Mrs. Mosquito.

“Well, I live now,” said Mrs. Hessian Fly, “and I am interesting enough for any use. I don’t make galls like so many flies, but simply lay my eggs in young blades of wheat, and when my little red babies hatch, they have only to crawl down and fasten themselves to the tender stalk, just below the ground. Don’t they love the sap, though? A field of wheat looks pretty sick after they have worked on it a while. Sometimes the wheat midges help them and then it is good-bye to the wheat. Mrs. Wheat Midge, you know,

lays her eggs in the opening flower of the grain, and her babies eat the pollen and ovule. You may guess what happens then.”

“I think it is real horrid to do that,” said Ruth.

“And what do you know about it, pray?” retorted Mrs. Hessian Fly. “We must all eat to live.”

“We certainly must,” said a house fly, flitting up with a loud buzz. “I have just escaped with my life. A cook wanted to take it because I tried to lay some eggs on her meat. What better place could a fly ask, I’d like to know? If Mrs. Blow Fly had been there, she would have put her eggs on that meat, screen or no screen. She is a most determined body and she can drop her eggs through the finest mesh, if she makes up her mind to do it.”

“Is Mrs. Blow Fly that big, buzzing, blue-bodied thing that is such a botheration?” asked Ruth.

“She’s big and blue, and she buzzes, or talks, with her wings, as we all do,” answered Mrs. House Fly, with dignity, “but she isn’t a thing. She’s a fly. There are hundreds of different kinds of flies, I’d like you to understand. The kind like me live in houses, but some prefer stables. They seem to like to stay with horses and cows, and are rather common. They have beautiful eyes, though, and plenty of them. Would you believe it, my head is nearly all eyes? I have thousands of tiny ones in my two big ones, not to mention the three single ones at the top of my head.”

“Gracious!” said Ruth. “No wonder it is so hard to catch you. But doesn’t it make you dizzy when you walk upside down, and how do you keep from falling?”

“Of course we don’t get dizzy and it is easy enough to keep from falling if you have pads and fine hairs on your feet. They just hold you to the place you are standing on. Men seem to consider this quite a wonderful thing. One of them has written some poetry about it. This is how it goes:

“What a wonderful fellow is Mr. Fly,
He goes where he pleases, low or high,
And can walk just as well with his feet to the sky
As I can on the floor.”

“Say,” spoke up a slim, narrow-winged creature with abnormally long legs, “I’m one of your relations, though I can’t walk upside down.”

“You?” repeated Mrs. House Fly, contemptuously. “Why, you can’t walk decently right side up.”

“It is true,” sighed the crane fly. “I haven’t even the grace of Daddy Long Legs, for:

“My six long legs all here and there
Oppress my bosom with despair.”

“Well, I don’t care about your legs,” said Mrs. House Fly. “I was speaking of my relations—my *smart* relations. All are not smart. I have some who need only bite the twig of a tree and lay their eggs there, and what do you suppose happens? A round ball grows over the spot and men call it a gall, but it is really a tiny house for my cousin’s babies. I have another cousin, whose name is *Cecidomyia strobiloides*. It is long for such a tiny creature, but she bears up very well under it.”

“I couldn’t ever pronounce it,” said Ruth. “What does she do, please?”

“She flies to a willow tree in the Spring, before the leaves are out, and with a spear on the end of her body she cuts a gash in the tip end of the bud, just where it is most tender and juicy. She lays an egg in the gash; then goes to another twig, and does the same thing, until she has laid as many eggs as she wishes. When her babies hatch, they do not look at all like their gauzy-winged little gray mother, nor do they care for sun or air. In fact, they never stir from their cells. They can eat, though, and the sap of the tree is their food.”

“You all seem to think a good deal of eating,” said Ruth.

“Of course. Isn’t that what we are hatched for? But my cousin’s babies have lost their appetites by the Fall, and then they go to sleep. They wake up in the Spring, and, strange to say, they have grown exactly like their mother and are ready to lay eggs on some more willow twigs. Very likely the willow tree does not care to have them do it, for the twig where their cradle is does not grow into a branch as the tree meant it should. Instead, the small leaves just crowd upon each other, until they look like a green pine cone.”

“I hope it will never happen to my willow tree,” said Ruth; “but please tell me more things. They are very interesting.”

“Interesting? I should say so. Indeed, I could go on talking all day, and not tell you one half the things we can do. But life is too uncertain to waste

it all in talking.”

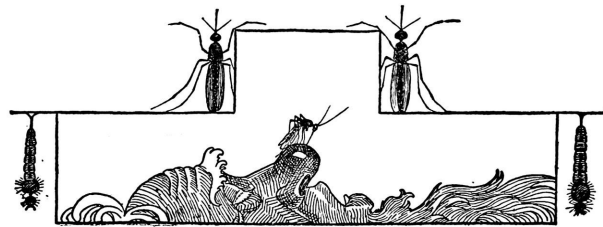
“Life is certainly full of accidents,” buzzed a big horse fly. “I’m here to tell Mrs. Mosquito, if she is looking for the messenger she sent out a while ago, she’d better make up her mind never to see her again. She went too near a horrid warty toad, and you can guess the rest.”

“We can,” sighed Mrs. Mosquito. “If it isn’t frogs, it’s toads and——”

“Often it’s birds,” finished Mrs. Horse Fly, “and they are the worst of all.”

“Such subjects remind me that I am hungry,” said Mrs. Mosquito, “and I’m off to find a juicy somebody for dinner. I think I shall lay some eggs too.”

“I wonder if it was my toad who ate that mosquito,” thought Ruth, as she watched the audience fly away.





CHAPTER V

RUTH HEARS ABOUT SOME WATER BABIES

An inner impulse rent the veil
Of his old husk, from head to tail
Came out clear plates of sapphire mail.

—Tennyson.

Ruth lay in the grass, under the old willow tree, watching a dainty little creature with a pale green body and four gauzy wings flashing with all the tints of the rainbow.

“What a beautiful dragon fly,” she said, half under her breath. “I never saw one so lovely before. I wonder if it is a dragon fly. Do you think it is, Belinda?”

“I am not a dragon fly,” came in answer from the dainty creature herself. “I’m a lacewing. Why don’t you use your eyes? It’s about time you learned something.”

“I do want to learn,” said Ruth meekly. “I am trying all the time. I wish you would tell me things. I thought you were prettier than most dragon flies.”

Mrs. Lacewing looked pleased. “Now you show your taste,” she said, “and I am quite willing to help you. Just wait a little while, and see what happens. Then if you don’t like it, well——” And without waiting to say more, or to let Ruth thank her, she was off.

“I think she means to come back,” said Ruth, expecting, she scarcely knew what, “and it will be nice, I am sure. Oh, Belinda, isn’t it just like living in Fairyland, since we can hear what they talk about? There! what did I tell you! It is Fairyland.”

Ruth added this with a rapturous little squeeze, for just then she saw the lacewing flying toward her, and with her many other beautiful winged creatures.

“The order Neuroptera, or the nerve wings,” said the lacewing, flitting close to Ruth, “that is some of them.” Then she introduced Ruth as a friend, adding in a self-satisfied tone: “She thinks I’m beautiful, and I quite agree with her, don’t you?”

Apparently the audience did. Of course she *was* beautiful, and, besides, she carried a scent bag which was not at all pleasant, and they knew they were likely to have the full benefit of it if they contradicted or displeased her.

“Now we’ll begin,” she went on, with the air of one who had settled all difficulties, but the next second she stopped, and, looking at a group of caddice flies, she asked sternly:

“Why are you here? and bless my wings, if there aren’t dragon flies, and stone flies, and, who would believe it, May flies. Now you know that not one of you belongs to our order.”

“Well, we belonged to it once,” answered a caddice fly, speaking for all.

“But I don’t understand,” began Ruth.

“Then don’t say anything,” put in a dragon fly, darting before her. “Keep quiet and listen, and you’ll learn things. Besides, it is very rude to interrupt people.”

Ruth felt snubbed, and tried to turn her back on the dragon fly, but, as he seemed to be everywhere at once, she found it impossible.

The caddice fly was still speaking. “We can’t always remember,” she said, “and I should like to know what right the wise men have to take us out of one order and put us in a sub-order.”

“Right is the last thing they think about,” spoke up a stone fly, “but I really care very little whether I’m called Neuroptera, as I was once, or Plecoptera, as I am now. Life is just as uncertain and full of accidents. Why, my friends, it is the greatest wonder I lived to grow up.” She sighed and began to fan her long, fat body with her broad fore wings.

“You know I was once a water baby.”

“Water baby?” repeated Ruth. “Wouldn’t your wings——”

“No they wouldn’t,” said Mrs. Stone Fly, “because I hadn’t any wings then. I was homely, flat, six-legged, and just the colour of the stone under which I spent most of my young life, hiding. I had to hide, or the boys would have found me and used me for bait. Think of it! Bait!”

And Mrs. Stone Fly, quite overcome, could say no more.

“We came to make a few remarks,” said one of a swarm of May flies that had been hovering about, “but we must go now. Life is too short for talking.”

“Poor things,” said Mrs. Lacewing, “life with them is indeed short. No wonder they are called Ephemerida. Think of living only for a day!”

“But they lived a long time as Nymphs,” said the dragon fly, who was still darting about, now here, now there, like a flash of living flame. “I know, because they were water babies like me. They could eat too, then, and the number of times they changed their skins was a caution. Why, my friends, they even change them after they leave the water and have their wings. No other insect does that.”

“Now, my story, in the beginning, is something like theirs. I, too, was born in the bottom of the pond and, no doubt, I played with some of you, or I may have tried to make a meal of you. Well, if I did I failed, and I shouldn’t be blamed for the sins of my youth. All of us eat when we can get the chance, and there’s no use in being sorry for the dinner. I suppose you would like to hear how I managed to get into the pond?” He looked at Ruth, who nodded her head, though she was still laughing at the idea of being sorry for a dinner.

“You see,” explained Mrs. Lacewing, “the dinner might be your nearest relation.”

“Just so,” agreed the dragon fly. “Now my mother, for of course I had a mother, though like most pond people I never knew her——”

“Do get to the point,” said an ant lion impatiently; “we are all growing old.”

“Well, the point is my mother,” answered the dragon fly, undisturbed, “but first I should say that I no longer belong to the order Neuroptera, but to the sub-order Ordonata. It means something about a tooth, but if I have any teeth, I don’t know it. Now to get back to the point: my mother flew down to the water one day, and when she left it there was a cluster of small yellow

eggs floating on the surface. I came from one of those eggs, and I didn't look like a dragon fly, I can tell you. I had six tiny spider-like legs, but not a sign of wings, and when I breathed it was not as I do now, like all perfect insects, through openings on each side of my body. I had gills, and a tube at the end of my body brought fresh water to them. This tube was a funny affair. It really helped me along, for when I spurted water through it I was pushed forward. Then I had a wonderful mouth, with a long under lip, that I could dart out and catch anything within reach, while I did not need to move my body at all."

"Just like frogs and toads!" cried Ruth.

"Not at all," answered the dragon fly. "They only send out their tongues. I send out my whole under lip. If you could only keep quiet you would not show your ignorance so plainly."

Once more Ruth was snubbed, and the dragon fly continued:

"In time I became a pupa."

Ruth looked the question she dared not ask.

"I'll explain," said the dragon fly, amiably. "Larva—that's what I was at first—means mask, or something that hides you. You will find out in time, if you do not know now, that the larva of an insect is really a mask which hides its true form. The plural of the word is larvæ. Now pupa, plural pupæ, means baby. It is usually the state of sleep in which the larva lies after spinning its cocoon or cradle, but in my case it didn't suit at all. Dragon flies, far from sleeping in the pupa state, seem to grow more active, and their appetites are larger. Indeed, I will say right here, everything that came my way, and was not too big, went into my mouth. In fact, I finally reached my limit and burst."

"Gracious!" cried Ruth in a shocked tone. "How *did* you get yourself together again?"

"Well, you see, the whole of me didn't burst. I simply grew too big for my skin, or my pupa case, as the wise men call it, and it cracked right open. I was climbing on a water plant when this happened, for all at once I had felt a longing to leave the water and get to the open air. My first effort was to get rid of the useless old shell which still clung to me, but I had quite a tussle before I could do so, and afterward I was very weak and tired. But the result was worth all my labour, for I found myself with these four wings,

and the rest of my beautiful body, and I needed only to dry myself before sailing away on the wind, the swiftest thing on wings, and the most renowned mosquito killer on record. Of course, my legs aren't arranged for walking. Why should they be? All six of them go forward, as if they were reaching for something, and so they are, reaching for something to eat. Woe betide any insect I start after. I catch him every time. I ought to, for I have thousands of eyes, and I can fly forward, backward, or any old way. I never stop to eat my dinner either. I hold it, and eat it as I go. Now if I had time, I would tell you how the children of Japan make a holiday, and go out to catch us for pets, and how they sing pretty songs to us and——”

“It is about time you stopped,” interrupted Mrs. Ant Lion. “You have tried our patience long enough, and I mean to speak this very minute. I’ve been told I am much like the dragon flies,” she added to the company, “but my babies are not at all like theirs. They do not belong to the water, and I am glad of it. I’m tired of water babies. I’ve heard so much of them to-day. My mother had the good sense to lay her eggs in sand, and I shall do the same. I was hungry from the minute I was hatched, and I would have run after something to eat right away, only I found I couldn’t. My legs were fixed in such a way I had to walk backward.”

“Backward?” echoed Ruth.

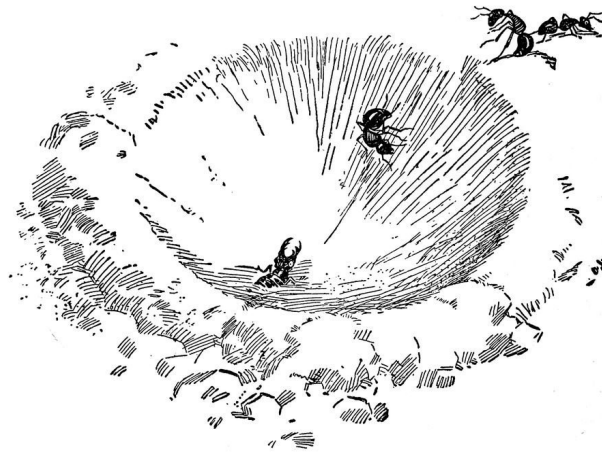
“Yes, backward. So there was nothing to do but to dig a trap for my dinner, and I set about it pretty quick. No one showed me how, either. I simply used my shovel-shaped head, and before long I had made quite a pit, broad and rounded at the top, and sloping to a point like a funnel at the bottom. You have seen them, of course?”

“I think I have,” answered Ruth.

“They are not hard to find if you keep your eyes open,” went on the ant lion.

“Well, as I said, I made one of these pits, and in the funnel end I lay in wait for ants. Soon one came along, slipped over the edge, as I expected, and tumbled right into my open mouth. Nor was she the only one. Some were strong enough to turn, even while they were slipping, and start to crawl up again, but I just heaped some sand on my head and threw it at them, and down they would come. My aim was always good, so were the ants, though I only sucked their juice. Of course I did not leave their skins around to frighten away other ants. I piled them on my head, and gave them

a toss, which sent them some distance away. After a time I stopped eating, and made a cocoon. Then I went to sleep!—for many days—during which I changed wonderfully, as any one must know who has seen ant lion babies and now sees me. This is all of my story, and I suppose we will hear about another tiresome water baby.”



“I MADE ONE OF THESE PITS AND IN THE FUNNEL END I LAY IN WAIT FOR ANTS”

“You shall hear about a water baby,” replied Mrs. Caddice Fly, waving her antennæ by way of salute, “but tiresome will do for your own homely children. I will begin by saying that, with the accidents of life, it is a wonder that any of us are here. When we caddice flies were hatched we were soft, white, six-footed babies. We were called worms, though we were not worms. Think of it! Soft bodied, with not very strong legs, white, and living at the bottom of the pond. Could anything be worse? No wonder we seemed to do nothing at first but try to get away from things that wanted to eat us. I tell you, pond life is most exciting. After a while the front part of our bodies and our heads began to turn brown, and, as the rest of us was white, and seemed likely to stay so, we all decided to make a case or house to cover our white part. So we set to work and of bits of sticks, tiny stones, and broken shells, glued together with silk from our own bodies, we made these cases. True, many of us went down the throat of *Belostoma*, the giant water bug, before we had finished, but those of us who didn’t crawled into our little houses, locking ourselves in by two strong hooks which grew at the end of our bodies. We could move about, but of course we carried our houses with us and——”

“How ridiculous!” said Mrs. Ant Lion. “Why didn’t you stay still?”

“Because we didn’t wish to,” answered the caddice fly. “We had to eat, and we had to get away from those who wished to eat us. At last we went to sleep, after first spinning a veil of silk over our front and back doors. I can’t answer for the others, but when I awoke I tore open my silken door, threw aside my pupa skin, and found I had wings. Since then I have had a new life, but even that has its enemies, and one never knows what will happen.”

With which doleful saying Mrs. Caddice Fly sailed away to the pond to lay some eggs among the water plants.

“Dear me,” said Mrs. Lacewing, “we seem to need something cheerful after that. I am glad I never lived in the water, if it makes one so blue. Now I shall tell you what my babies *will do*, not what I *have done*. Of course it is the same thing, but it is looking forward rather than to the past. After this meeting is over I shall lay some eggs, on just what plant I haven’t yet decided, but it will be in the midst of a herd of aphides. Be sure of that. Aphides are plant lice,” she explained, seeing the question in Ruth’s eyes. “You will learn more of them later. Now as to the way I shall lay my eggs: First, from the tip of my body I shall drop a thick gummy fluid, and draw it out into a long, stiff, upright thread, and upon the end of this thread I shall fasten an egg. I shall lay a number of eggs in this way, each on its own pole, so to speak. Some people may think my way odd, but it is very wise. A lacewing knows her children. They are not beautiful. Such short-legged, spindle-shaped things couldn’t be pretty, but they are sturdy, and they have an endless appetite.”

“I should think they would feel lonely on those ridiculous poles,” said Mrs. Ant Lion.

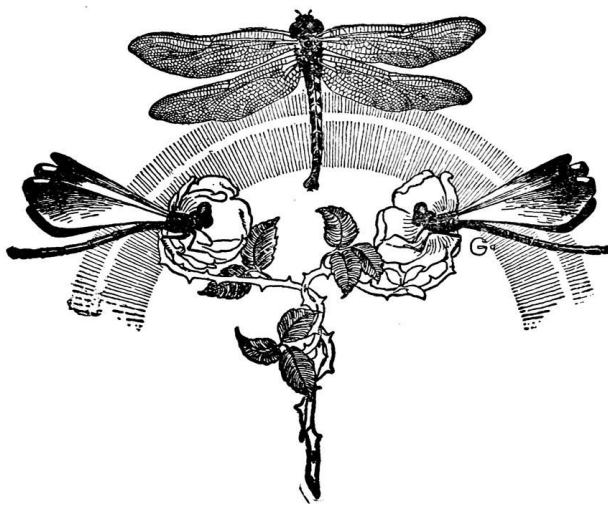
“Not at all. They are not there long enough to feel lonely. They are in too great a hurry for dinner. They are hungry, with a big H. Now just suppose I should lay my eggs as the rest of you do, ever so many together, what do you think would happen? I will tell you in a few words. The dear child who came out first would eat all his unhatched brothers and sisters. He doesn’t, only because he can’t reach them.”

“It’s a wonder he doesn’t eat his pole,” said Ruth, her face showing what she thought of such babies.

“Yes, it is,” agreed Mrs. Lacewing, “but, strange to say, he doesn’t seem to care for it. Indeed, he leaves it as quickly as he can, and goes hunting. Of

course he needn't hunt far, for he is in the midst of aphides. Every mother looks out for that, and really it is quite a pleasure to see him suck the juice from aphid after aphid, holding each one high in the air in his own funny way. So you can see why lacewing babies are friends to the farmer and the fruit grower, for aphides kill plants and trees, and young lacewings kill aphides. They can eat and eat and eat, and never grow tired of aphides. Indeed, they really deserve their name—aphislion. When they do stop eating it is to fall into their long sleep, but first they weave a cocoon as beautiful as a seed pearl, in which they change into a most lovely creature—one like me. Now our meeting is adjourned, and I hope a certain person has learned a few things."

"Oh, ever and ever so many, thank you," answered Ruth gratefully.





CHAPTER VI

RUTH GOES TO A CONCERT

Oh, sweet and tiny cousins that belong,
One to the fields, the other to the hearth,
Both have your sunshine.

—Leigh Hunt.

Ruth and Belinda were crossing the meadow, when a big grasshopper made a flying leap, and landed on Belinda's head.

"Do excuse me," he said; "I missed my aim. No one hurt, I hope, or frightened?"

"Oh, no," answered Ruth. "Belinda is real sensible; she isn't afraid of anything, and I am just as glad—as *glad*—to see you. Maybe you will——"

Ruth hesitated, hoping he would know what she meant to say. She was sure he could tell her a great many things, if only he would. He was so polite and nice; besides, he looked very wise.

"I suppose you're going to the concert," said Mr. Grasshopper, after waiting a second for Ruth to finish her sentence.

"Concert?" she repeated, opening her eyes wide. "What concert?"

"Why the Straightwings' Concert. They give one every sunny day in Summer. Didn't you know that? Dear me, where were you hatched and where have you been living since? Well, why do you stare at me so? Don't you like my looks?"

"Oh, yes," Ruth hastened to answer. "You look very nice—something like a little old man."

"I've heard that before, and there's a story about it. Shall I tell it?"

"Yes, please; I just love stories."

“Very well. Once upon a time, long, long ago, there lived in Greece a beautiful young man named Tithonus. Now it chanced that Tithonus loved Aurora, the Goddess of the Dawn.”

“Greece?” said Ruth. “Why, that’s where Arachna lived, the one who turned into a spider, you know?”

“Do you want to hear my story or don’t you?” asked Mr. Grasshopper, sharply.

“I do want to hear it. I really do.”

“Very well, then, don’t interrupt me again. As I was saying, Tithonus loved Aurora, and every morning he would lie in the meadow and wait for her coming. Then the fair goddess would give him her sweetest smiles. But one day Tithonus grew pale and ill, and all the love of Aurora could not make him well again. ‘Alas!’ he cried, ‘I am mortal, and I must die.’ ‘Nay,’ answered Aurora, ‘you shall not die, for I will win for you the gift of the gods.’ And, speeding to the mighty Jupiter, she begged that Tithonus might be as a god, and live forever. So for a while they were happy together, but as the years passed Tithonus grew old and bent, for Aurora had forgotten to ask that he might always be young. Grieving much, Tithonus lay under the shadow of the trees and sighed through the long days.”

“‘Ah, my Tithonus,’ whispered Aurora, ‘I love you too well to see you thus unhappy. No more shall you be sad or bend beneath an old man’s weakness, but, as a child of the meadow, happy and free, you shall sing and dance through the golden hours.’ In that moment Tithonus became a grasshopper, and ever since then his descendants have danced and sung in the sunshine. That’s the end of the story. I might have made it twice as long, but Summer is so short, and I want to dance.”

“It was a very nice story,” said Ruth, “but do you really dance?”

“Of course, our kind of dancing.”

“But don’t you do lots of other things too?”

“Yes; we give concerts, and we eat. We are hatched with big appetites, and a strong pair of jaws, and we start right in to use them on the tender grasses around us. We only follow our instincts, though men call it doing damage. You eat, don’t you?”

“Why, yes, but I don’t eat grass, you know.”

“Because it isn’t your food. You see it’s this way: In the kingdom of nature all creatures have a certain work to do, and each is exactly fitted for its place, for all are governed by laws more wonderful than any man has made. Not that I wish to speak lightly of man, he is good enough in his place, but he is apt to think himself the whole thing, and he isn’t. Maybe he doesn’t know that for every human creature on earth there are millions of plants and animals.”

“Oh,” said Ruth, “really and truly?”

“Really and truly. You couldn’t begin to count them, and do you know, if the earth was to grow quite bare, with only one living plant left on it, the seeds from that one plant could make it green again in a very few years. But if certain insects were left without other creatures to eat and keep them down, the poor old earth would soon be bare once more. So you see there must be laws to fix all these things. Nature balances one set of creatures against the other, so there will not be too many of any kind.”

Ruth had listened in open-eyed astonishment. Surely this was a very wise grasshopper.

“You know a great deal,” she managed to say at last.

“Yes, I do,” was the answer. “I heard two men say the things I’ve just told you. They were walking across this meadow, and I listened and remembered. You see, I believe in learning even from men. But do listen to the concert—we are right in the middle of it.”



THE WISE
GRASSHOPPER

They certainly *were* in the middle of it. The zip, zip, zip, zee-e-ee-e of the meadow grasshoppers seemed to come from every part of the sunny field, while the shorthorns, or flying locusts, were gently fiddling under the grass

blades, their wing covers serving for strings, and their thighs as fiddle bows, and the field crickets, not to be outdone, were scraping away with the finely notched veins of the fore wings upon their hind wings.

The longhorns were also there, some in green, others in brown or gray, all drumming away on the drum heads set in their fore wings.

“You would hear katydid too,” said Mr. Grasshopper, “only he refuses to sing in the day. He hides under the leaves of the trees while it is light, and comes out at night. If you think *me* wise, I don’t know what you would say of him. He is such a solemn-looking chap, always dressed in green, and his wing covers are like leaves. You might think him afraid if you saw him wave his long antennæ, but he isn’t. He is curious, that’s all. It is a high sort of curiosity, too, like mine—a wish to learn. I suppose you know we don’t make our music with our mouths?” he asked suddenly. “Well, that is something,” he added, as Ruth nodded “Yes.”

“I sing with the upper part of my wing covers, but my cousins, the shorthorns, sing with their hind legs. Why do you laugh? Aren’t legs as good to sing with as anything else?”

“I—I suppose so,” said Ruth. “It sounds funny, because I am not used to that kind of singing.”

“Just it. Now I shall tell you a few more facts about us. We belong to the order of the Straightwings, or the Orthoptera, as the wise men call it.”

“Will you please tell me what that means?” asked Ruth. “Do all insects belong to something ending in *tera*? Most everything I have talked to does except toads and spiders.”

“And they are not insects,” said Mr. Grasshopper. “Not even the spiders. The word insect means cut into parts, and all insects have three parts, a head, and behind that the thorax or chest, and the abdomen. Then, too, they always have six jointed legs. Now maybe you have noticed that spiders are not built on this plan? There are only two parts of them. The head and thorax are in one. It is called the cephalothorax. I’d feel dreadfully carrying such a thing around with me, but the spiders do not seem to mind it. Their other part is their abdomen. I heard a little boy say it was like a squashy bag; and between ourselves that is about what it is. Of course you know that spiders have eight legs and that alone would settle the question. True insects never have but six. Now as to the orders: All insects are divided into groups, and it is something about the wings which gives them their names.

That is why they all end in ptera, because ptera comes from pteron, a word which means wing. It isn't an English word, you know, but is taken from a language called Greek."

Ruth listened very patiently. If she had heard all this in school it would have seemed very dry, but when a grasshopper is telling you things it is of course quite different.

"But I am sure I can never remember it all," she said.

"Ah, yes, you can. Remembering is easy if you only practise it."

"Why, that's like the White Queen," cried Ruth. "She practised believing things till she could believe six impossible things at once, before breakfast."

"I don't know the person," said the grasshopper.

"She lived in the Looking Glass Country," began Ruth, but Mr. Grasshopper was not listening.

"You have met the Diptera, or Two Wings," he said. "That's easy. Then you've met the Neuroptera, or Nerve Wings. That's easy too. And now you have met the Orthoptera, or Straightwings, meaning me, and if I'm not easy, I should like to know who is. You see our wings are——"

"Wings?" said Ruth in surprise.

"Of course. Look here," and opening his straight wing covers, Mr. Grasshopper showed as nice a pair of wings as one could wish to possess. "Not all of us have wings," he added, folding his own away, "but those of us who have not live under stones. Our order includes graspers, walkers, runners, and jumpers. Not all are musicians. The graspers live only in hot countries. Maybe you have seen the picture of one of them—the praying mantis he is called, just because he holds up his front legs as if he were praying. But it isn't prayers he is saying. He is waiting for some insect to come near enough so he may grab and eat it. That will do for him. Next come the walkers. The walking stick is one, and he isn't a good walker either, but the stick part of the name fits him. He is dreadfully thin. There is one on that twig now, and he looks so much like the twig you can scarcely tell which is which."

"Why, so he does," said Ruth, poking her finger at the twig Mr. Grasshopper pointed out. "Isn't he funny?"

"Indeed," grumbled the walking stick. "Maybe you think it polite to come staring at a fellow, and sticking your finger at him, and then call him

funny, but I don't. I want to look like a twig. That's why I am holding myself so stiff. I have a cousin in the Tropics who has wings just like leaves."

"Yes," added the grasshopper, "and his wife is so careless she just drops her eggs from the tree to the ground and never cares how they fall."

"Well, if that suits her no one else need object," snapped the walking stick. "I believe in each one minding his own business."

"An excellent idea," said Mr. Grasshopper. "Now let me see, where was I? Oh! the runners; but you'll excuse me, I will not speak of them at all. They include croton bugs and cock roaches, and it is quite enough to mention their names. With the jumpers it is different. They are the most important members of the order. I'm a jumper, I am also a true grasshopper. You can tell that by my long slender antennæ, longer than my body. For that reason I am called a longhorn, but my antennæ are really not horns."

"I don't see how any one *could* call them horns," said Ruth.

"No more do I, but some people have queer ideas about things. Well, I don't care much. There is my mate over there. Do you notice the sword-shaped ovipositor at the end of her body? She uses it to make holes in the ground and also to lay her eggs in the hole after it is finished. Yes, she is very careful. Her eggs stay there all Winter, and hatch in the Spring, not into grubs or caterpillars, or anything of that sort. They will be grasshoppers, small, it is true, and without wings, but true grasshoppers, which need only to grow and change their skins to be just like us. And I'm sure we have nothing to be ashamed of. We have plenty of eyes, six legs, and ears on our forelegs, not like you people who have queer things on the sides of your heads. Such a place for hearing! but every one to his taste. Well, to go on, we have wing covers, and lovely wings under them, a head full of lips and jaws, and a jump that *is* a jump. What more could one wish? Do you know what our family name is?"

Ruth didn't know they had a family name, so of course she could not say what it was.

"It is Locustidae," said Mr. Grasshopper, answering his own question. "Funny too, for there isn't a locust among us. Locusts are the shorthorned grasshoppers—that is, their antennæ are shorter than ours. They are cousins, but we are not proud of them. They are not very good."

“No one is asking you to be proud,” said a grasshopper, jumping from a nearby grass blade. She had a plump gray and green body, red legs, and brown wings, with a broad lemon-yellow band.

“What’s the matter with me?” she demanded. “I guess you don’t know what you are talking about. It’s the Western fellow that is so bad. We Eastern locusts are different.”

“Well, I suppose you are,” agreed the longhorn. “I know the Western locusts travel in swarms and eat every green thing in sight. They are called the hateful grasshoppers.”

“No one can say that our family has ever been called hateful or anything like it,” said a little cricket with a merry chirp. “We are considered very cheery company, and one of the sweetest stories ever written was about our English cousin, the house cricket.”

“I am sure you mean ‘The Cricket on the Hearth,’” said Ruth. “It is a lovely story, and I think crickets are just dear. Are you a house cricket too?”

“No, I belong to the fields, and I sing all day. Sometimes I go into the house when Winter comes and sing by the fire at night, but my real home is in the earth. I dig a hole in a sunny spot and Mrs. Cricket lays her eggs at the bottom, and fastens them to the ground with a kind of glue. Sometimes there are three hundred of them, and you can imagine what a lively family they are when they hatch.”

“I should like to see them,” said Ruth, for it was quite impossible for her to imagine so many baby crickets together.

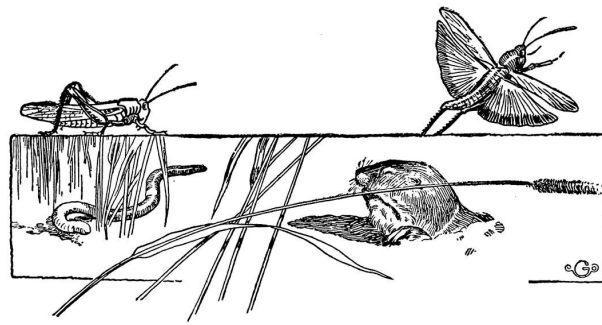
“Well, it is a sight, I assure you,” answered the little cricket. “Did you ever come across my cousin the mole cricket? She is very large and quite clever. She makes a wonderful home with many halls around her nest. She is always on guard too so that no one may touch her precious eggs. Then I have another cousin, who doesn’t dress in brown like me, but is all white. He lives on trees and shrubs and doesn’t eat leaves and grass as we do. He prefers aphides. You can hear him making music on Summer evenings. We crickets seldom fly. We——”

The sentence was not finished, for just then a long droning note grew on the air, increasing in volume, until it rose above the meadow chorus.

“Oh!” cried Ruth, spying a creature with great bulging eyes and beautiful, transparent wings, glittering with rainbow tints, “There’s a locust!

Isn't he beautiful, Belinda? Maybe he will tell us some things. Oh, Belinda, aren't we in luck?"





CHAPTER VII

RUTH MEETS MANY SORTS AND CONDITIONS

The shrill cicadas, people of the pine,
Make their summer lives one ceaseless song.

—Byron.

“**A** locust, indeed,” said the newcomer, and Ruth could see plainly that he was not pleased. “It does seem to me you should know better than that. Can’t you see I have a *sucking* beak and not a *biting* one, like the grasshopper tribe? Besides, my music isn’t made like theirs. No faint, fiddly squeak for me, but a fine sound of drums.”

“I think I’ll move on,” said Mr. Grasshopper, and Ruth could see that he was quite angry. She turned to look at the cricket, but he was far across the field, fiddling to his mate.

“I wish you wouldn’t go,” she said to the grasshopper. “You have been so nice to me and I have learned ever so much from you.”

“Oh, I dare say,” was the answer. “More than you will learn from some people I could mention, but I really must leave you. My mate wants me.” And a flying leap carried him quite away.

“There, we are rid of the old grandfather,” said the cicada, “and now what can I do for you?”

“Tell me your real name if it is not locust,” answered Ruth.

“It certainly is not locust. I’ve been called a harvest fly, though I am not a fly either. I’m a cicada, and nothing else, and I belong to the order of bugs.”

“And what kind of *tera* is it?”

“*Tera*?” repeated the cicada, looking at her with his big eyes. “Oh, yes, yes, I understand. You mean our scientific name. It is Hemiptera, meaning

half-wings. I know we have some objectionable members, but I don't have to associate with them, and I rarely mention their names. I have a cousin who lives in the ground seventeen years. Think of it! Of course he is only a grub and doesn't care for air and sun. I lived there two years myself, but I was a grub also then. You see my mother put her eggs in the twig of a tree, and when I came out of one of them I wanted to get to the ground more than I wanted anything else, so I just crawled out to the end of the branch and let go. Down I went, over and over, to the ground, where I soon bored my way in, and began to suck the juices of the roots about me. I liked it then, but I couldn't stand it now. Of course the moles were trying. They were always hungry and we were one of the things they liked for dinner. One day something seemed to call me to the world of light, and I came out a changed being—in fact, the beautiful creature you see before you now. Perhaps you do not know how much attention we have attracted? In all ages poets have sung of us, even from the days of Homer. Maybe you will not believe me, but the early Greeks thought us almost divine, and when Homer wished to say the nicest things about his orators he compared them to cicadas. A while ago I told you we were sometimes called harvest flies. We have also been given the name Lyremen. Shall I tell you why?"

"A story!" cried Ruth, clapping her hands. "Oh, yes, please tell it!"

"Very well. Once upon a time, ages ago, a young Grecian player was competing for a prize, and so sweet was the music he drew from his lyre that all who heard it felt he must surely win. But alas! when he was nearly finished one of his strings snapped, and, with a sad heart, he thought that all his hope was gone. Not so, however, for a cicada, drawn from the woods by the sweet sounds, had perched upon the lyre and when the musician's trembling fingers touched the broken string it gave forth a note that was clear and true. Thus again and again the cicada answered in tones that were sweet and full. When the happy player realized that the cicada had won the prize for him, he was so filled with gratitude that he caused a full figure of himself to be carved in marble, and in his hand a lyre with a cicada perched upon it. Now wouldn't you be proud if your family had such a nice story about them?"

"I'm sure it is very nice," agreed Ruth.

"Yet I'm not one to brag," added the cicada, "and I am never ashamed to say I'm a bug. Now if you will come with me to the pond I will show you

some of my cousins. They are very interesting.”

And with a whiz the gauzy-winged fellow darted up into the sunshine, and Ruth, following him across the meadow, could only hug Belinda in a rapture of expectation, and whisper in a low voice:

“Aren’t we in luck, Belinda—just the best kind of luck?”

They had gone only a little way, however, when a mole pushed his strong little snout above the ground.

“Gracious! what a noise,” he said. “If I had had a chance when you were a baby you wouldn’t be here now to disturb quiet-minded people.”

Ruth jumped. She thought the mole meant he would have eaten her. Then she laughed. “Of course it was the cicada he was talking to,” but the cicada didn’t mind.

“I know that very well,” he answered, cheerfully, “but you didn’t get me. That makes all the difference, and now you can’t.”

“Well, nobody wants you now. You would be mighty dry eating, but when you were a grub, oh, my! so fat and juicy, like all the other grubs and slugs and worms. I eat you all. Yet what thanks do I get from man for doing away with so many of his enemies? Complaints, nothing but complaints, and just because I raise a few ridges in the ground. I can’t help that. When I move underground I push the earth before me, and, as it has to go somewhere, it rises up.”

“What do you push with?” asked Ruth, sitting down in front of the mole.

“With my snout and forepaws,” he answered, “what else? The muscle which moves my head is very powerful, and you can see how broad my forepaws are, and, also, that they turn outward. They help to throw back the earth as I make my way forward. I have ever so many sharp little teeth, too, and my fur lies smooth in all directions, so it never rumples and——”

“Do come on,” interrupted the cicada; “that fellow isn’t interesting.”

“That’s so,” said a thin little voice, as an earthworm cautiously lifted his head from the ground. “Has he gone?” he asked anxiously. “He’d eat me sooner than wink if he saw me. It is warm and damp this morning, that is why I am so near the surface. I don’t like dry or cold weather. My house _____”

“Have you a house?”

Ruth had turned upon him in a second, full of questions as usual.

“Certainly I have a house. It is a row of halls, lined with glue from my own body. The walls are so firm they can’t fall in. Underground is really a delightful place to live, snug and soft, cool in Summer, warm in Winter. Lots to see, too. All the creeping, twining roots and stems reaching out for food, storing it away, or sending it up as sap to the leaves. The seeds waking up in the Spring, and hosts of meadow and wood people wrapped in egg and cocoon, who spend their baby days there. Quite a little world, I assure you. Of course I can’t see any of these things. I have no eyes.”

“Oh!” said Ruth, “how dreadful!”

“No, it is just as well. If I had eyes I might get earth in them. I go through the ground so much.”

“But isn’t that awful hard work?” asked Ruth, shutting her eyes to realize what having no eyes might mean.

“It isn’t hard when one has a nice set of bristles, as I have to help me along.” The earthworm was one who saw the best side of everything. “I am made up of more than a hundred rings,” he went on, “and on each are small stiff hair-like bristles so, though I have neither eyes, ears, hands, nor feet, I am quite independent. I can move very fast, and the slime that covers me keeps the earth from sticking to me. Do you know I am the only jointed animal that has red blood? It is so. I do no harm, either, to growing things, and I help to build the world. My tunnels let air into the ground and help to keep it loose. I also bring up rich soil from below, and lay it on the surface. I also——”

“Well, that’s enough,” interrupted the cicada, moving his wings impatiently. “I thought you wanted to see *my* relations?” he added to Ruth.

“So I do,” answered Ruth. “Where are they?”

“There are a number of them right in this meadow, though you would never think it, to look at them. They are not at all like me. See that white froth clinging to those grass stems? A cousin made that. Of the sap of the plant too. If you look, you will find her in the midst of it. She is green and speckled and very small. Then there are the tree hoppers, as funny in shape as brownies, and the leaf hoppers. They are all my cousins. The aphides too. Of course you know the aphides?”

“I believe they were the things Mrs. Lacewing told me I should learn about later,” said Ruth, with sudden remembrance.

“Very likely. Mrs. Lacewing’s children should know about them. The aphides are very bad, though they are so very tiny. But what they lack in size they make up in numbers. Really there are millions of them. They are not travellers, either, but stay just where they are hatched, and suck, suck, suck. In that way they kill many plants, for it is the sap of the plant, its life juice, which serves them for food. They eat so much of this that their bodies can’t hold it all, and what they don’t need is given off as honey dew. The ants like this honey so well that to get it they take good care of the aphides. But there are some aphides which do not give off honey dew. Do you see this white stuff on the alder bushes?”

“Yes. I’ve often seen it before, too. It looks like soft white fringe.”

“Well, it isn’t. It is a lot of aphides, each with a tuft of wool on its body, and a beak fast stuck in the alder stem.”

They had now reached the pond, which lay smiling in the sunshine.

“It would be so pretty,” said Ruth, throwing herself down on the grass, “if it wasn’t for the horrid, green, oozy stuff all over it.”

“Horrid, green, oozy stuff?” repeated the cicada. “Child, you don’t know what you are talking about. That green stuff is made up of tiny green plants more than you could count. Each has a rootlet hanging down like a silver thread and leaves almost too small to be called so. They are green though and they do the mighty work of all green leaves, for, besides shading the pond world from the hot rays of the sun, they make for the many inhabitants the life-giving oxygen without which they would die. And I want to tell you something more: In that duckweed—for what you call green, oozy stuff is duckweed—there are millions of tiny living things too small to be seen by the eye except with the aid of a microscope.”

Ruth looked quite as astonished as the cicada meant she should be.

“You have a great deal to learn, I assure you. Maybe you haven’t thought of the pond as a world, but just see what a busy place it is.”

Ruth looked and agreed with the cicada. Dragon flies were darting here, there, and everywhere; frogs, with their heads out of the water, seemed to be admiring the scenery when they were not swallowing air or whatever else came in their way; glancing minnows and bright-eyed tadpoles played

amongst the swaying water weeds; even the wrigglers were there, standing on their heads in their own funny way; and the water striders, skating after their own queer fashion. Yes, it was a busy place.

A party of whirligig beetles came dashing by, circling, curving, spinning, and making such a disturbance that a backswimmer lost his patience and told them to be quiet.

They didn't like that at all, so they threw about him a very disagreeable milky fluid which made the backswimmer dive for the bottom in a hurry.

"That settled him," said one of the whirligigs. "Hello! friend Skipper Jack," he called to a water strider, "what are you doing?"

"Skating, of course," answered the water strider. "There, they are gone," he added, to the cicada, "and I am glad of it. They are nuisances."

"You are right," agreed the cicada.

"I am glad they don't belong to our order."

"Don't they?" asked Ruth. "I think they are awfully funny."

"Funny or not, they are beetles," answered the water strider. "You had better use your eyes. Do you know why I can skate and not get my feet wet? No, of course you don't, and yet it is as plain as the nose on your face. I have a coat of hairs on the under side of my body. That's why. I spend my time on the surface of the water, for my dinner is right here. Plenty of gnats, insect eggs, and other eatables. Then if I wish I can spring up in the air for the things that fly. My Winters I spend under water, but for other seasons give me the surface."

"And I like the bottom best," said a water boatman, showing himself quite suddenly, his air-covered body glittering like silver armour.

"Another cousin," whispered the cicada in Ruth's ear. "He is called the water cicada, as well as water boatman."

"He looks more like a boat than he does like you," said Ruth.

"My body is boat-shaped," spoke up the boatman; "and see my hind legs; they really are like oars, aren't they?"

"I am wondering what brought you to the surface," said the cicada.

"Why, I let go my hold on that old water weed, and you know the air that covers my body makes it lighter than the water and unless I cling to something I naturally rise. It is inconvenient, for I do not need to come to

the surface for air. I can breathe the same air over and over, because I know how to purify it.”

“How do you do it?” asked Ruth. Surely these insects were wonderfully clever.

“Oh, I simply hang to something with my front legs, while I move my back ones just as I do in swimming, and that makes a current of water pass over my coat of air and purify it. That fellow swimming on his back over there is obliged to come to the surface every little while. He carries air down in a bubble under his wings.”

“Do you mean me?” asked the backswimmer, making a sudden leap in the air, and flying away.

“Gracious!” cried Ruth in surprise. “I didn’t know he could fly.”

“There’s a good deal you don’t know,” replied the water boatman, a remark Ruth had heard before. “I can fly too,” and he also spread his wings and was off.

“Well,” said the cicada, “I guess we might as well be off too. There seems to be no one in sight to interest us.”

“What about cousin Belostoma?” asked a sort of muffled voice, as a great pair of bulging eyes showed themselves above the water, and out came the giant water bug as big as life.

“I’ve just had my dinner,” he said. “It really is funny to see how everything hides when Belostoma shows his face. My wife is the only one who doesn’t seem to be afraid of me and she—well, she’s a terror and no mistake.”

“Why, what’s the matter now?” asked the cicada.

“And what has happened to your back?” added Ruth, with eager curiosity.

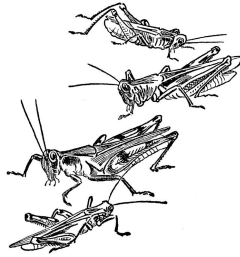
“My wife’s happened, that’s what,” answered Belostoma in a doleful tone. “She laid her eggs a while ago and glued every blessed one to my back. It is nothing to laugh at either. There’s no joke in being a walking incubator. Well, I must be going now. It is dinner time.”

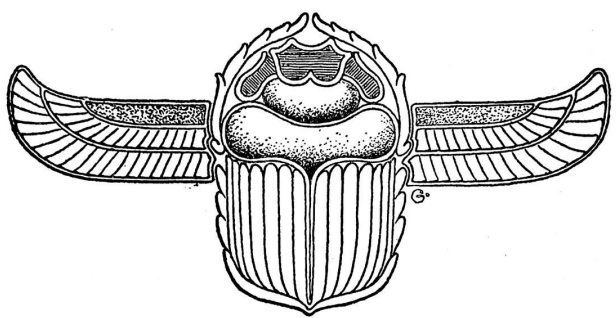
“I thought you just had your dinner,” said Ruth.

“Yes, but it’s time again. It is always time. How silly you are.”

“I must go too,” said the cicada, “but it isn’t dinner that calls me. I feel sure my mate is longing for some music and I’m off to give her a bit. See you later.”

And, spreading his wings, the cicada flew away, beating his drums as he went.





CHAPTER VIII

MRS. TUMBLE BUG AND OTHERS

Their wings with azure green
And purple glossed.

—*Anna L. Barbauld.*

Something exciting was going on. Ruth could not tell just what it was at first. She could only watch and wonder. Then her eyes grew large and bright. Surely some fairy's wand had touched the old orchard, for suddenly it seemed alive with beetles—big beetles and little beetles; beetles in sober colourings, and beetles gleaming with all the tints of the rainbow. Ruth had never dreamed that there could be so many of them or that they were so beautiful.

The gorgeously coloured, graceful tigers attracted her first, though she didn't know their name.

"Oh," she cried, "how lovely!"

"And how strange," added a voice just above her head, "how very strange, their children should be so homely."

"What's that?" asked one of the tigers, a metallic green fellow, with purple lights, and two pale yellow dots on the edge of each wing cover. "Our children not so beautiful as we are, did you say? Of course, they are not; a fat grub couldn't be, you know. But let me tell you, there are few things as smart as a tiger beetle baby. I say," he added, looking full at Ruth, "have you ever seen the hole he digs? It is often a foot deep, while he is less than an inch long. He has only his jaws and fore legs to work with too. Yet he piles the earth on his flat head as if it were the easiest thing in the world, and then, climbing to the top, he throws it off, and is ready for another load."

“I suppose he digs a hole to catch things,” said Ruth, “like the ant lion, and does he stay at the bottom and——”

“No, he doesn’t stay at the bottom. He watches near the top of his hole for his dinner, hanging on by a pair of hooks which grow out of a hump on his back. He always goes to the bottom to eat his dinner, though; he seems to like privacy. Yes, we are a fierce family from the beginning, for we grown tigers can catch our prey either running or flying, and we usually manage to get it, too. But, then, farmers need not complain of us, for we never eat plants, and that is more than can be said of many here.”

“Such taste,” said a cloaked, knotty horn, holding herself in a position that showed off her changeable blue and green dress, and her short yellow cape.

But the tiger did not answer. He was off after his dinner. Several tree borers, however, nodded their heads in agreement.

“I believe in a vegetable diet myself,” said Mrs. Sawyer, who wore as usual her dress of brown and gray. “It is just such people as the tigers who make things like that necessary in a respectable meeting,” and as she spoke she waved her very long antennæ toward a big sign which read:

“THE AUDIENCE ARE REQUESTED NOT TO EAT EACH OTHER DURING THE MEETING”

“I am glad to say I am not one of that kind. I wonder if any one of you know why the members of our family are called sawyers. Perhaps I had better tell you: It is because our children saw into the trunks of evergreen trees, and sometimes they make holes large enough to kill the trees. Smart, isn’t it, for a baby?”

“But it doesn’t seem to be very nice,” began Ruth. Then she stopped, for Mrs. Sawyer was looking at her and the borers were nodding their heads again.

“Our children do not saw,” said the borers, “but they do bore, and it is pretty much the same thing for the tree.”

“My friends,” broke in a very solemn voice.

Every beetle stopped talking, and Ruth jumped to her feet, then flopped down on the grass again, waiting for what was coming.

The speaker, a large, clean-looking beetle, had just flown to a twig in the very middle of the meeting. He was black in colour, well sprinkled above

and below with pale straw yellow in dots and points, but the queer thing about him was the two oval velvety black spots, each with a narrow line of straw colour around it, on his thorax. They were like great eyes, and made him look very wise.

“He is the eyed-elater,” whispered Mrs. Sawyer to Ruth. “There he is speaking again.”

“My friends,” the big beetle was saying in tones as solemn, as before, “the important thing in any meeting is to keep to the main issue.”

“The main issue?” said the goldsmith beetle, a beautiful little creature with wing covers of golden yellow, and a body of metallic green covered with white, woolly fuzz. “What is the main issue?”

“Dinner,” replied the tiger beetle, returning to his old place. “If it isn’t breakfast or supper.”

“No, my friend,” said the eyed-elater, with a grave glance, “the main issue is——”

Then he stopped and fixed his two real eyes and the two spots which looked like eyes on some small beetles which were leaping in the air, turning somersaults, and making quite a noise.

“Will you be still?” he said in his sternest voice.

“How foolish,” said Mrs. Sawyer, “to expect click beetles to be still!”

But Ruth was all curiosity.

“I’ve seen you before,” she said, going closer and touching one of the funny little fellows.

Suddenly it curled up its legs, dropped as if shot, then lay like one dead.

“Here, here!” called the elater. “No more of that! We know all about your tricks!”

“All right,” said the would-be dead one, and he gave a click, popped into the air several inches, and came down on his back.

“That won’t do at all,” he said, and, clicking and popping once more, he came down on his feet.

“There,” he added, “you need to have patience with click beetles. You ought to know that, friend elater, for you are one of us.”

“Well, I’m bigger, and not so foolish, and my children are not so harmful as yours. Think of being a parent of those dreadful wire worms! That is

what you click beetles are, and you know the farmer hasn't a worse enemy. Now we must get back to the main issue."

"Back?" said Mrs. Sawyer. "Were we ever there to begin with? You can't scare me," she added, "no matter how hard you stare. You haven't any more eyes than the rest of us. Those two spots are not real eyes, and you know it."

"The main issue," repeated the elater in a very loud voice, "is, What makes us beetles?"

"That's something I'd like to know," said a handsome little beetle in a striped coat. "I'm a beetle, if there ever was one, yet I have a world-wide reputation as a bug."

"Pray don't get excited, Mrs. Potato Bug. It isn't your time to talk yet. We are on the main issue, and I will answer my own question."

Ruth was glad some one would answer it, for at this rate it seemed they would never get anywhere.

"We are beetles for several reasons," went on the elater. "In the first place, we belong to the order Coleoptera."

Another tera, thought Ruth.

"That name is taken from a language called Greek, and means sheath wing. It is given to us because we have handsome outside wings which we use to cover our real flying wings. All beetles have them, though those of our cousin, Mr. Rove Beetle, are quite short."

"That's a fact," said a rove beetle, "and no one need think we have outgrown our coats. It is simply a fashion in our family to wear our sheath wings short. We can always fold our true wings under them, and I'd like to see the fellow who says we can't."

"Well, you needn't get so mad about it," answered the elater in mild tones.

"And don't curl your body up as if you were a wasp," added Mrs. Sawyer. "Everybody knows you can't sting."

"I don't care," said the rove beetle. "I hate to be misunderstood. We are useful too. I heard a man call us scavengers. I don't know what it means, but something good, I am sure, from the way he said it. I must be going

soon. It is so dry here. You know my home is in damp places under stones or leaves.”

“You may go when you wish,” answered the elater. “We are still on the main issue. As I said before, we are beetles, and there is no reason to take us for bugs. Calm yourself, Mrs. Potato Bug. We have no sucking beak as the bugs have, but we have two sets of horny jaws, which move sideways, and *not* up and down. These are to bite roots, stems, and leaves of plants, so most of our order live on vegetable food and are enemies to the farmer, but some of us are his friends, for we eat the insects that injure his crops. Our children are called grubs. Some of them make a sort of glue, with which they stick together earth or bits of wood for a cocoon; others make tunnels in tree trunks or wood and transform in them. We may well be proud, for we belong to a large and beautiful order, and we are found in all parts of the world. We are divided into two sub-orders—true beetles and snout beetles. I hope our cousins, the snout beetles, will not be offended. They are real in a way.”

“The farmer and fruit grower think so anyway,” said a little weevil. “We have been called bugs just because we have a snout, but any one can see at a glance that it isn’t a bug’s snout. It is not a tube at all, but has tiny jaws at the tip.”

“I don’t believe I could see all that,” said Ruth rather timidly, for these clever little people had a way of making her feel she knew very little.

“Maybe you can’t,” was the short answer, “and I dare say you can’t tell how we use our snouts either. We punch holes with them in plums, peaches, cherries, and other fruits, not to mention nuts and the bark of trees. I am a peach curculio, but that is not important. We all work in the same way—that is, drop an egg in the hole made by our snout, then use the snout again to push the egg down. Mrs. Plum Weevil is busy now in the plum orchard back of us; so of course she couldn’t come to this meeting. ‘Duty before pleasure,’ she said. She will lay eggs in quite a number of plums, and the plums will drop from the trees before they are ripe.”

“And there’ll be a lump of gum on them!” cried Ruth, clapping her hands.

The weevil looked at her with approval. “You do notice some things,” she said.

“The gum oozes out of the hole made by our snouts. Of course our egg hatches inside the fruit, and the baby has its dinner all around it. As it hasn’t a leg to walk on——”

“Dear! dear!” sighed the elater. “You seem to forget that we are trying to keep to the main issue. As I said before——”

“You are always saying what you said before,” snapped Mrs. Sawyer.

“Now, they are beginning again,” thought Ruth, but the elater paid no attention to Mrs. Sawyer.

“As I said before,” he repeated, “we have reason to be proud, for though we build no cities, like ants, wasps, and bees, and make no honey or wax, or have, in fact, any special trades, yet we are interesting and beautiful. The ancient Egyptians thought some of us sacred and worshipped us.”

“There!” cried Mrs. Tumble Bug, literally tumbling into their midst. “I couldn’t come at a better time.”

Ruth gave a little scream of delight when she saw her, and Mrs. Tumble Bug nodded with the air of an old friend.

As usual, her black dress looked neat and clean, though she and her husband had rolled and tumbled all over the road in their effort to get their ball to what they considered the best place for it. They had succeeded, and Mrs. Tumble Bug’s shovel-shaped face wore a broad smile in consequence.

“I knew about this meeting,” she said, “but my husband and I agreed that duty should come before pleasure.”

“She heard me say that,” whispered the little peach weevil to her nearest neighbour.

“I didn’t,” answered Mrs. Tumble Bug. “I have just come. We only found a safe place for our ball a little while ago.”

“That ball!” said Mrs. Sawyer in disgusted tones. “I should think you would be tired of it.”

“Tired of our ball?” repeated Mrs. Tumble Bug. “Why, our ball is the most important thing in the world. This was a big one, too. We made it in Farmer Brown’s barnyard, and then I laid my eggs in it, and we rolled it all the way here. Of course it grew on the road, and I couldn’t have moved it alone, but my mate helped me. He always helps. Indeed it seems to me

tumble bugs are the only husbands in the insect world who care about their children's future."

"Now I know," said Ruth, who had been thinking very hard. "You think so much of your balls because they hold your eggs. I've often wondered about them."

"Of course that is the reason," answered Mrs. Tumble Bug; "and when our eggs hatch the babies will have a feast all around them."

"Ugh!" said Ruth, and some flower beetles shook their little heads, and added:

"It would be better to starve than eat the stuff in that ball."

"Tastes differ," said Mrs. Tumble Bug, amiably; "but, speaking of sacred beetles, it was our family the Egyptians worshipped. They could not understand why we were always rolling our ball, so they looked upon us as divine in some way, and made pictures of us in stone and precious gems. They can be seen to-day, I am told, but I do not care about that. I must make another ball," and, nodding to her mate, they left the meeting together.

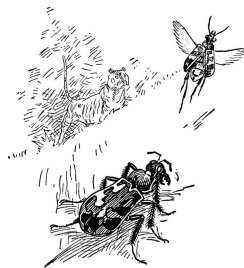
"Now we'll adjourn for dinner," announced the elater, much to the disgust of Mrs. Potato Bug, who was just getting ready to speak.

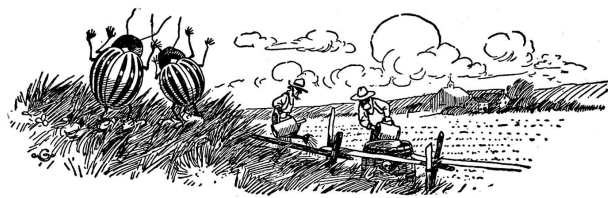
"Dinner is well enough," she said, "but how is one to enjoy it when one must stop in a little while?"

"You needn't stop," answered the elater. "Stay with your dinner. We are not so anxious to hear you talk."

"But I mean to talk, and I *will*," and Mrs. Potato Bug was off to the potato field, intending, as she said, to take a light lunch, and be back when the meeting opened.

But potato bugs propose, and farmers dispose, and——





CHAPTER IX

LITTLE MISCHIEF MAKERS

It's a wonder, it's a wonder
That they live to tell the tale.

—Anon.

Mrs. Potato Bug did not return. A sister bug rose to speak when the meeting opened after dinner. There had been a sad tragedy in the potato field, she told them, and even at that very minute the farmer and the farmer's men, armed with barrels of "pizens," were waging a warfare in which millions of potato bugs were going down to their death. "Alas! my friends," she finished with a sigh that seemed to come from the very tips of her six feet, "no words can paint the dreadful scene. She who was here but a short while ago, so chipper and so gay, even she was giving her last gasp as I fled from the field of carnage."

The story moved the audience deeply, and all agreed that something should be done to suppress the farmers. It was even suggested to appoint a committee to consider ways and means, but at this point a very young potato bug asked the question:

"If there were no farmers, who would plant potatoes for us?"

"No one," answered Mrs. Sawyer, who was there just as self-important as ever. "Then maybe there would be no potato bugs, and I for one wouldn't be sorry."

"Indeed," said the potato bug who had told the tale of battle, "I'd have you know we are Colorado beetles, if you please, and our family has a world-wide fame. We are true Americans, too, and not emigrants from Europe, like many other insects, and that reminds me: The other day when I was having a nice chew on some very juicy potato leaves, I heard

somebody say to somebody else: ‘Oh, young Lochinvar is come out of the West.’ He said a lot more, but I heard that plainly, and I wondered if he meant our family, and didn’t know our name, because, you know, we came out of the West.”

“I am sure he didn’t mean you,” said Ruth, who was in her old place right in the middle of the meeting. “That line is from a lovely piece of poetry about——”

“No one asked your opinion,” answered the potato bug angrily. “It is bad enough to have outsiders force themselves in, without being obliged to hear their silly remarks.”

Ruth’s face grew red, and she was about to reply, when Mrs. Sawyer whispered in her ear.

“Don’t mind her, she is only a potato bug.”

It was well that Mrs. Potato Bug did not hear this. “Before 1859,” she was saying, “our home was in the shade of the Rocky Mountains. There we fed on sandspur, a plant belonging to the potato family, and the East knew us not. It was only after the white settlers came West and planted potatoes that we found out how much nicer a potato leaf is than a sandspur leaf, so of course we ate potato leaves. We came East, travelling from patch to patch, and by 1874 we had conquered the country to the Atlantic Ocean. That shows what a smart family we must be, and I will tell you how we do. We lay our eggs on the potato leaves, and our children find their dinner all ready, and, as they hatch with splendid appetites, they get right to work. Those that hatch in the Fall sleep all Winter in the ground and come out as beetles in the Spring, just in time to lay more eggs. So we keep things going, especially the potatoes.” And Mrs. Potato Bug retired with the air of one quite proud of herself.

Her place was taken by a little ladybug, looking quite pretty in her reddish-brown dress, daintily spotted with black.

“I have several cousins,” she said, “of different colours, but all spotted and all friends to farmers and fruit growers, for we eat the aphides and scale bugs which do so much harm to plants. We are called bugs, but of course we are beetles. I could tell you a story——”

“Never mind the story,” said a great brown blundering fellow, much to Ruth’s regret, for she wanted to hear the story.

“Excuse my awkwardness,” said the newcomer. “It bothers me to fly by day. I like to go around the evening lamps. I can buzz loud enough for a fellow three inches long, though I am really not one. I am called a June bug, and I’m really a May beetle. What do you think of that? I have been told that the farmers do not like us, nor our children either. They are such nice, fat, white grubs too. They do love to suck the roots of plants though, and, as we grown fellows are just as fond of the leaves, between us we make the poor old plants pretty sick.”

“I wish something had made you sick before you came here to disturb quiet folks with your buzzing,” said a large blue beetle, dropping some oil from her joints in her excitement.

“Oh, it doesn’t matter,” she added when Ruth spoke to her about it. “It only proves that I have a right to be called an oil beetle. In these days it is so important to know who is who.”

Ruth was watching the oozing oil curiously.

“Does it hurt?” she asked.

“Oh, no,” was the answer. “It is perfectly natural. I can’t move about fast, I am too fat, and I haven’t any wings to speak of. So when anything disturbs me I can only play ’possum and drop oil. I wasn’t always like this, though,” she went on, with a heavy sigh. “Would you believe it? I was born under a stone in a field of buttercups. I was tiny, but my body had thirteen joints and three pairs of as active little legs as you ever saw. Each had a claw on it too. What do you think of that? I used my legs right away to climb a nearby flower stalk. Something inside of me seemed to tell me just what to do, and when a bee came flying by, though she looked like a giant, I wasn’t a bit afraid, but I popped on her back, and clutched so tight with my six little claw-like legs she couldn’t have gotten me off if she had tried. But maybe she didn’t know I was there. Anyway, I had some lovely free rides, for she flew from flower to flower, and then she went home.”

“Oh,” interrupted Ruth, “did you go right into the hive?”

“Yes, but I didn’t notice much about it at first. I felt very tired, and I can only remember dropping from her back and going to sleep. When I awoke a funny thing had happened.”

“What?” asked Ruth, full of curiosity.

“My legs were gone, and only a half dozen short feelers were left me instead. But I didn’t mind. I was in one of the tiny rooms of the hive, and there was a nice fat bee baby for me to eat. I didn’t lose any time either; I was hungry. Besides the baby there were bee bread and honey. Who could ask for more? Indeed, I ate so much I went to sleep again, and, would you believe me? in that sleep I lost even my short feelers, and, worst of all, my mouth.”

“Gracious!” said Ruth.

“I suppose after that I slept again, for what’s the use of staying awake if you can’t eat? But that nap finished me. I waked up looking as I do now. It was a sad change. Maybe that is why I feel so blue and am called the indigo beetle.”

“I don’t see why you changed so many times,” said Ruth.

“Neither do I. No other insect does, but I suppose it has to be. I shall soon lay my eggs, and that no doubt will be the end of me. We seem to begin and end with eggs.”

She sighed heavily, and went on: “I have a cousin who is used to make blisters on people. Think of it! She is called Spanish fly, and she is no more a fly than you are.”

“Does she bite them to make the blister?” asked Ruth.

“Dear me, no! The poor thing is dried and made into powder and then spread with ointment on a cloth. That makes the blister. I suppose it takes ever so many of my poor cousins for just one blister. I tell you, life is sad.”

“Do stop that sort of thing, I can’t stand it!” said a plain, slender little beetle, with no pretensions to beauty of any sort. “I came here as a special favour, and then I am forced to hear such talk as that. I am never at my best in the day, and you should know it. Some of you complain of being called bug, and others object to the name fly. Now I am as much a beetle as any of you, and I’ve been called both bug and fly.”

“A lightning bug?” cried Ruth.

“Yes, and also firefly, and if it was dark I’d prove it. Of course my light can’t be seen in the day, and generally I’m not to be seen either, for we fireflies hide away on the leaves of plants until it begins to grow dark. Then we come out, and have gay times flying over the meadows. Some of our family who live in warm climates are so large and bright they are used to

read by. Not only that, ladies wear them as they would jewels, and in Japan _____”

But the firefly could say no more, for just at this moment some whirligig beetles came flying in and every one turned to look at them.

“I should like to know what those fellows are doing here,” said a bumble-bee beetle, making such a loud humming that Mrs. Sawyer declared she thought a real bumble bee was in their midst. “People who live in the water shouldn’t belong to our family, anyhow. I can’t imagine any one liking the water.”

“That’s because you are not a water beetle,” answered one of the whirligigs.

“Why, the water is the most sociable place in the world. Something lively happening all the time. Constant changes too. Those who are with us one moment are gone the next, but that is life on land as well as in the water for us insects. Dinner is always our first thought. Of course we water fellows are fitted for our life. We are put together more tightly than you land beetles, and we are boat-shaped besides. We use our hind legs for paddles, and we have wings with which we can leave the water if we wish. We whirligigs are sociable fellows, always a lot of us together, and such fun as we have dancing and whirling about in the water! We don’t often dive unless something is after us.”

“You must have very good times,” said Ruth, watching the shiny, bluish black little beetles with eager attention. Then she asked quite suddenly:

“Have you four eyes?”

“No, my dear,” answered the first speaker, “we have only two. They look like four, because they are divided into upper and lower halves. So you see we can look up and down at the same time, and, I tell you, insects need to step lively to keep out of our way. Good times? I should say we did have good times. Now to the surface to snatch bubbles of air with the tiny hairs on the tip of our tails, and then down again for a race or a game of tag with our friends. No, not all the water beetles are as frisky as we are. Some are—now what is that?”

The whirligig might well ask the question, for a sound like a tiny popgun had broken in upon his remarks, and the whole audience, including Ruth of course, was looking at a greenish blue beetle who had just come in, leaving

a fine trail of smoke behind him. It was he who had made the queer noise, and he seemed quite disturbed by the sensation he was creating.

“Do excuse me,” he begged. “I really forgot I was among friends.”

“I should think so,” answered the elater, looking at him sternly. “A beetle who carries a gun should be careful about using it.”

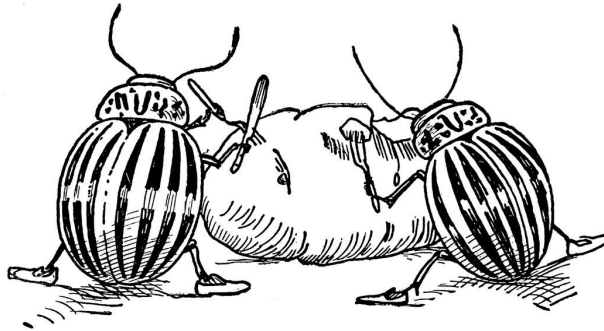
“Well, I try to be careful, but accidents will happen.”

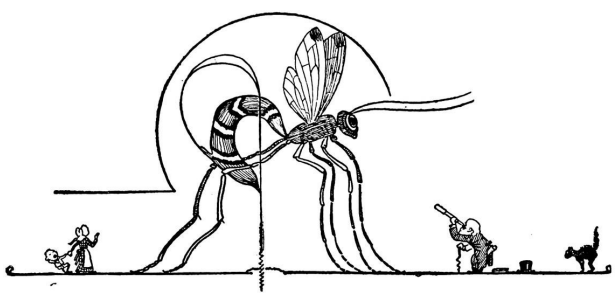
“Yes, you might really call it a gun,” he said, in answer to Ruth’s question, “and I have been named the Bombardier beetle because I carry it. When men try to catch me, I shoot it off, though I suppose it really doesn’t hurt them, but it quite blinds my insect enemies until I can get away, anyhow. Oh, no, I do not use balls or shot. It is a fluid, in a sac at the end of my body, and when I spurt it out it turns to gas, and looks like smoke.”

“Well, we have had talk enough for to-day,” interrupted the elater, and the Bombardier beetle said no more.

“Talk?” repeated Mrs. Sawyer, “I should say so. Very tiresome talk too. Now I’m going out to lay some eggs. I know a lovely tree.”

“That’s all she thinks about,” said the elater. “I’m sure we have had a very interesting meeting, and I made the main issue very plain.”





CHAPTER X

SOME QUEER LITTLE PEOPLE

That nothing walks with aimless feet.

—*Tennyson.*

In a corner of the garden, where the lilacs grew tall and broad, Ruth was waiting for something to happen. She had a feeling, as she told Belinda, that the most interesting things were coming, for the wind had been kissing her cheeks and ruffling her hair, just as though it was saying to her, “Watch now. Watch closely and listen.” Then, too, the garden seemed to be alive. Bees droning over the flowers; wasps collecting their tiny balls of wood pulp or marketing for their families; ants running here, there, and everywhere; not to mention many other winged creatures, some of whom were made after a fashion so queer that Ruth, forgetting how rude it is to make personal remarks, deliberately asked of one:

“If you please, what is that long piece which seems to be growing from the tip of your body? It looks like Mary’s stove hook when she sticks it in the lid.”

“That,” was the rather short answer, “is my abdomen, and it isn’t growing from the tip of my body, but from the *top* of my thorax. It seems to me you have never seen an ensign fly before.”

“No, I never did. Please, what does ensign mean?”

“The dictionary will tell you that. All I know is some man got an idea that we carried our abdomens aloft like a flag or ensign, and so named us ensign fly. We are not flies, to begin with, but we have to keep any idiotic name they choose to tack on us. Now take Mrs. Horntail, who wants——”

“Thank you, I can speak for myself,” interrupted the horntail, sharply. She was quite handsome, with her black abdomen banded with yellow, her red and black head, yellow legs and horn, and dusky wings.

“I like my name. It means something, for I have a horn on my tail, and, what’s more, I use it. You should see me bore into solid green wood. None of your dead wood for me. I am not content with one hole either. I bore a great many, and in each I drop an egg, and when my babies hatch they get fat on the sap wood of the tree.”

“There seem to be such a lot of things to eat trees,” said Ruth.

“Perhaps there are, but I am interested in horntail babies only. They do their share of eating too, and when they grow sleepy they make cocoons of chips and silk from their own bodies, and go to sleep. After they wake they are changed into winged creatures, who naturally do not care to live in the tree any more. So they gnaw their way through the bark to the outside world and——”

“Not if the woodpeckers and I can help it,” interrupted an ichneumon fly, keeping her antennæ in constant motion. She seemed to have long streamers floating from the back of her, and, altogether, Ruth thought her even queerer looking than the ensign fly.

“Those streamers are my ovipositor,” she explained to Ruth. “The thing I lay eggs with, you understand. When I shut them together they form a sort of auger, with which I bore into a tree, way, way in, where the fat horntail babies are chewing the sap wood, and so ruining the tree. Into their soft bodies I lay my eggs and when my children hatch they eat, not the tree, but the horntail baby. It is a wonderfully good riddance, and so the farmer and fruit grower consider us their friends and call us ‘trackers,’ because we find the hiding places of so many pests that harm the plants.”

“You can’t get my babies,” said Mrs. Saw Fly. “I haven’t a horn, but I have a saw, and, though it will not bore into wood, it saws fine gashes in green leaves. Of course I drop an egg in each gash, and soon there’s a swelling all around it, and when my children hatch they rock in gall nut cradles, and the sap which gathers there is their food.”

“Talk about gall cradles,” said a gall fly, “my sisters and I are the fairies who make them to perfection. Each of us has a different plant or tree which she prefers, and each follows her own fashion in making galls, and we

puzzle even the wise men. Have you ever seen the brown galls that grow on oaks?"

"Why, of course," answered Ruth, glad the question was such an easy one.

"Well that's something, but I doubt if you have noticed the rosy coloured sponge that sometimes grows around the stem, or the mimic branch of currants drooping from the spot where the tree intended an acorn to be, or the tiny red apple-like ball on the leaf."

Ruth shook her head. "They must be very pretty," she said.

"Pretty? I should say so. They are all different kinds of galls too, and we gall flies make them. Sometimes we sting the leaf, sometimes the twig, and sometimes the stem, and always just the kind of cradle we intended grows from it, and the egg we laid there hatched into a baby grub, ready to eat the sap."

"Then you know about the one on the willow tree," put in Ruth. "The one the housefly told about. It grows like a pine cone, and is made by some one with a dreadfully long name."

"That is something entirely different," answered the gall fly. "We do not pretend to make all the galls, you understand. Some are made by insects belonging to quite another order. The willow tree cone is one. You may always know ours from the fact that we make no door for the babies to come out, as other insects do. Our babies make their own door when they are ready to leave their cradle. And now to show how much is in some names, I will tell you that those other gall insects are called gall gnats and belong to the order of flies, while we are called gall *flies*, and belong to the order Hymenoptera."

"Oh!" cried Ruth, clapping her hands. "Now I know the kind of tera you belong to, Hy-men-op-tera," she repeated slowly. "Please tell me just what it means."

"No, I won't," was the ungracious answer. "I hate explanations."

"I'll tell you," said Mrs. Horntail. "I know all about it." And as Ruth turned to her with grateful eyes she began:

"Hymenoptera means membrane wing, and that's the kind we have, though some of our order have no wings at all. The others have four wings, the front pair being larger, with a fold along the hind edge, that catches on

hooks on the front edge of the hind wings; so we really seem to have but one pair. Do you understand that?"

"Yes," nodded Ruth.

"Very well. We are divided into two sub-orders: stingers and borers. Our larvæ are called maggots. They are not like us, being white grubs, with round horny heads, pointed tails, six legs——"

"Here, here!" said the ichneumon fly, "that does well enough for your children, but you know perfectly well that the babies of the rest of us have no legs."

"Yes, I know. Poor things! Legless children! How sad! Mrs. Saw Fly and I are the only exceptions."

"And your children use their legs to no good purpose either," said the ichneumon fly.

"My children need no legs. They never move from the spot where they are hatched until after they transform. Why should they? Their dinner is right there."

"The same with mine," added a little bright-coloured brachnoid. "I choose a nice fat caterpillar, or something like that, to lay my eggs in, and he always lasts until my babies are ready to spin their cocoons, which they do on his shell, or dried skin, or whatever you choose to call it. I know he himself is quite gone. It is a pretty sight to see them."

The brachnoid herself was a pretty little thing and as she looked not unlike the ichneumon fly, only smaller, Ruth asked Mrs. Horntail if she were not a young ichneumon fly.

"Young ichneumon?" repeated Mrs. Horntail. "Whoever heard of such a thing? A young ichneumon is as large as an old one. None of us insects grow after we leave our cocoons. When we are what you mean by young—babies, in other words—we are different. I thought you had learned that before now. Haven't you had larvæ and pupæ explained to you?"

"Oh, yes," said Ruth, "but I had forgotten. Of course you are different when you are first hatched, and then you get wings, while you sleep, but I thought maybe you grew even after you had wings."

"Some of the grasshopper tribe do that, and spiders are hatched little spiders and grow bigger as they grow older, but we do no such thing. Besides, as you heard a while ago, an ichneumon baby is legless, absolutely

legless, and homely. Well, I think the homeliest thing that lives, but then what can you expect with such a mother?"

"I don't think she is so awfully homely," said Ruth. "She is odd-looking, and—and——"

"Odd-looking?" repeated Mrs. Horntail. "You should see her drilling a hole and laying her eggs. If she doesn't cut a figure, I don't know one. With her abdomen all in a hump, her wings sticking straight up, and her antennæ standing out in front, not to mention the ridiculous loop she makes with the ovipositor, she certainly is a sight."

"But I find the horntail babies," said the ichneumon fly, quite undisturbed, "and that is the important thing. I wonder if this meeting is over?"

"I hope so," answered Mrs. Horntail. "It is not a proper meeting at all. If I had the regulating of it, I would make some of these creatures behave. See that ant on the pebble over there. She is making faces, actually making faces."

"I am not making faces," answered the ant. "I am getting ready to talk, and I haven't had a chance."

She was little and brown, and scarcely an eighth of an inch long, but she looked quite important as she prepared to address the audience.





CHAPTER XI

WISE FRIENDS AND FIERY ONES

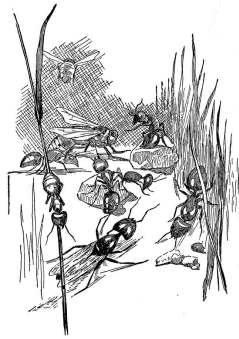
A was an ant, who seldom stood still,
And who made a nice nest in the side of a hill.

—*Edward Lear.*

“Sh!” said Ruth to the audience in general, for she wanted very much to hear what the ant had to say. The ant looked at her approvingly, and then said in a very solemn tone:

“My friends, there are ants and ants.”

“Who doesn’t know that?” snapped Mrs. Horntail.



“MY FRIENDS,
THERE ARE
ANTS AND
ANTS”

“Yes, there are ants and ants,” repeated the speaker, not noticing the interruption. “There is the carpenter ant, for one. In the books she is called *Componotis Pennsylvanicus*, but never mind the name. It doesn’t seem to hurt her. She makes her nest in the trunks of trees, old buildings, logs, and

places of that kind. You can see her on the leaf by Mrs. Saw Fly. She is large and black and——”

“Clean,” finished the carpenter ant, speaking for herself, and, without asking further permission, she poised on her hind legs and began to ply her tongue, and the fine and coarse combs on her fore legs, until she had gone over her whole body, smoothing out ruffled hairs, and getting rid of every atom of soil. Her toilet done, she gave a few leisurely strokes, then drew her fore legs through her mouth to clean the combs, and stretched herself with an air of satisfaction.

“I hope I haven’t interrupted the proceedings,” she said, “but if I am not clean I am miserable. Now, Miss Lassius Brunens, please go on.”



““THEN THERE
ARE ANTS WHO
KEEP SLAVES””

“Miss who?” asked the little brown ant. “Oh, I see. You are calling me by the name the wise men give me. Well, I can stand it. To continue: I have mentioned the carpenter ant, and there are also the mound builders. Everybody knows their big hills. Then there are ants who keep slaves, and live under stones, and there are honey ants, who live in the South and use the abdomens of their own sisters to store honey in, and there are ants who sow seed and harvest it, and ants who cut pieces from green leaves and carry them as parasols, and soldier ants and——”

“Oh, give us a rest!” broke in Mrs. Horntail. “I am tired of ants.”

“Jealous, you mean,” said the little brown ant, “because you are not as wise as we are. Maybe you don’t know that whole books have been written about us and our clever doings. And men have spent years and years trying to study our ways. Now my family may not be the most wonderful, but I think it is the best known. We are the little ants who make the hill with a

hole in the middle, which you so often see on sandy paths, or roadsides, or in dry fields.”

Ruth had edged closer, and was listening eagerly. Once more the little ant looked at her approvingly, then went on:

“Some people think our houses are queer, because they are dark. Of course we have no windows, only a door, and that is a hole in the roof. We like it so though, and you might be surprised if you could see our many wonderful galleries and chambers. We made them all too. Dug them out of the earth, with our feet, throwing the soil out behind us, until the burrow grew too deep. Then we had to take it out grain by grain. We made our pillars and supports also, using damp earth for mortar. We don’t mind work, but we *do* mind human giants carelessly putting their feet in the middle of our hill and breaking in upon our private life. Those accidents will happen though, and our first thought is always the babies. They have no legs, and we have no hands, so we take them in our jaws, and speed away with them to our underground chambers, where they will be safe. I have seen human babies carried when they *did* have legs. There is no excuse for that.



“‘THEN THERE
ARE ANTS WHO
CUT PIECES
FROM GREEN
LEAVES AND
CARRY THEM AS
PARASOLS’”



THE HOUSE OF THE MOUND-BUILDER ANT

“Another thing, I know better than to call a human baby an egg, but, would you believe me, there are lots of people who think our babies are eggs. I have heard them called so. Now the reason we are so careful of our babies is because if there were no babies there would be no ants, and that brings me to the queen, for without her there would be no babies, because there would be no eggs, and babies always begin by being eggs. Only the

queen lays eggs, remember that. She is important for this reason, and no other. She is not our ruler, as some suppose. In fact, we have no ruler. Ants do as they please, but they usually please to do what is best for the whole community. We have many queens, but they are not jealous of each other, as the bee queens are. They do not look like us workers. They are ever so much larger, and were hatched with wings. The males also have wings, but it really matters very little what they have. They are such a weakly set, and after they go abroad with the queens, when they take the one flight of their lives, they usually die, or something eats them, and so they are settled. It is the queens who interest us. Some of them we never see again. They go off somewhere and start new colonies, or something may eat them too, but those that come back either unhook their wings, or we do it for them. Then they settle down and begin to lay eggs. Their egg laying is not after the fashion of bee queens, who go to certain cells and leave eggs in them. The ants drop their eggs as they walk around."

"Don't they get lost?" asked Ruth.

"No, indeed. Workers follow and pick up every one. They take good care of those precious eggs, too, and when they hatch into helpless grubs, without wings or feet, our work begins in earnest. Every morning we carry them into the sunshine, and bring them down again at night. We fondle them too, and keep them clean by licking them all over. Then of course they must be fed, and, like other babies, they prefer milk."

"And I know where you get the milk!" cried Ruth, all excitement. "It is from the aphides, isn't it? The cicada told me. The aphides are his cousins. He doesn't think so much of them, but he says you do."

"Well, why shouldn't we? They give us the most delicious milk. We have a fine herd of aphides now pasturing on a stalk of sweetbrier, and when Winter comes we will keep their eggs down in our nest, and put them on the sweetbrier in the Spring, so that the little aphides which hatch from them will have plenty to eat. Yes, and we may even build tiny sheds for them to keep their enemies from reaching them."

"I wonder if you intend to talk all day?" broke in a sharp voice. "I sha'n't wait another minute."

It was not Mrs. Horntail, as Ruth thought at first, but Madame Vespa Maculata, or, in plain English, the white-faced hornet, and, as she was a fiery lady, no one disputed her when she said:

“I am the largest and most distinguished of my family, and I build a nest whose delicacy and beauty make it a wonderful piece of insect architecture. It is proper that I should speak first, and I will speak right now.”

“Speak, by all means,” said the little ant. “I have quite finished.”

“Then move,” answered Vespa; “I need space.”

The whole audience gave it to her, including Ruth, who did not edge up close, as she did to the other speakers.

“It is this way,” she whispered to Belinda. “Those sharp people are very interesting, but it is better not to get too near until you know them quite well.”



“VESPA
MACULATA”

“I suppose,” Madame Vespa was saying, “I suppose we wasps can scarcely be called general favourites. We have a sting, you see, but, my friends, that was intended for laying eggs, and if we use it on people it is because they meddle in our business. It is our way. We *will* sting those who bother us. Now, in our community—for we are social wasps—the female is unquestionably the better half. We have our rights and we insist on them. My mate was a good-for-nothing fellow, like the rest of them. I didn’t marry him until Fall, and he soon left me, and did nothing but perch around in the sunshine with others like him, and I had all the hard work of the home. Finally he died. I suppose he couldn’t help that, but I doubt if he would have made an effort anyhow. Well, reproaches are of no use now, for he is very much dead by this time. I have had a whole Winter’s sleep since I saw him last. We queen wasps always sleep in Winter. We are the only ones of the colony who do not die when cold weather comes. You see, our community is not like the bees. It lasts only for a Summer, and each Spring the queens wake up and start a new one. That was what I did. I slept in the crevice of a barn and left it full of plans. You can imagine the task before me, but I was plucky and soon chose a tree to suit me. My house was made of paper, and I should like to say right here that we wasps are the first paper makers in the world, for while Egypt still traced her records in stone, or on the inner bark of the papyrus, my ancestors were manufacturing paper, that man has finally learned to make in the same way. For paper is only vegetable fibre reduced to a pulp and pressed into sheets.”

Ruth's eyes were wide with astonishment, and she was edging nearer to Madame Vespa.

"Can you really make paper out of wood?" she asked.

"Of course. See my jaws? They are made to chew wood. Not decayed wood either. That may do for wasps who live under ground, for the brownish paper it makes isn't strong enough to stand exposure. I choose good wood, and I make fine gray paper."

"I wish you would tell me how you do it," begged Ruth.

"Why, I simply gnaw the wood with my powerful jaws, and chew it until it is a pulpy mass, then I spread it in a sheet, wherever I wish it, and smooth and pat it with my feet. See how flat they are? I have heard of people beginning their houses at the cellar and building up. I consider that perfectly ridiculous. I always begin at the top. First, I make a slender stem or support to fasten the nest to the tree. Then I make three or more six-sided cells, which I hang from the support, and lay an egg in each, fastening it in with glue, for the open side of the cell is down. After this I enclose my cells with a wall of paper, and by this time, I am glad to say, my children begin to hatch, and though at first they look like horrid little worms, who can't help themselves at all, I always know they will grow like me soon, and do a great deal of work.

"Feeding them isn't an easy job, I can tell you, especially when it is added to my other duties, but, after a while, each baby weaves a little silken door over its cell, and goes to sleep. When she wakes she is a wasp, and the first thing she does is to wash her face and polish her antennæ, nor is it long before she gets to work. My first children are always workers, and after a number of them are hatched I can give my whole time to laying eggs."

"But when the nest is once done?" began Ruth, who had forgotten her fear entirely and was now quite close to Madame Vespa.

"The nest done?" repeated the fiery lady. "You should know that our nest is never done. New cells must be added, old walls gnawed down, and fresh ones built up to enclose larger combs. Indeed, we are never idle. We ventilate as the bees do, and we have sentinels too. Later in the season I lay eggs that hatch out drones, and last of all, the queen eggs. They are——"

"Now you would think," said a yellow jacket, buzzing up excitedly, "you would really think that Vespa might mention the fact that other wasps exist,

but not she. Now I want to tell you, the white-faced hornet *isn't* the whole thing. There are yellow jackets too."

"We have eyes," said Madame Vespa, "but go ahead and talk, and get through, for pity's sake."

"Yes, I mean to talk, and I shall get through when I please. We always insist that people shall respect our rights, and they generally do or—something happens. Our nests are quite as remarkable as Vespa's, though we do not hang them from trees, as she is in the habit of doing. Our cousin, Mrs. Polistes, also makes a paper nest, but she builds only a layer of cells, with not a sign of a wall about them. Any one can look right in on her private life."

"I'm quite willing they should," spoke up Mrs. Polistes, a long, slender brown wasp, with a yellow line around her body. "I could wall up my house if I wished to, but I *don't* and I *won't*; so there."

"They all have awful tempers, haven't they?" said Ruth to Mrs. Horntail.

"Tempers?" repeated that lady. "They are perfect pepper pots, though I must say Mrs. Polistes isn't usually as bad as the others."

"I am talking," called the yellow jacket, "and the rest of the audience will please keep still. As I was saying, though I doubt if you all heard it, there are other members of our family who have not been mentioned yet. We have miners, masons, and carpenters just like the bees. Of course they are solitary, and——"

"I object!" interrupted Mrs. Muddauber. "I won't be bunched in with ever so many others. I will speak for myself."

She was quite graceful, with a waist as slender as a thread, but she jerked her wings about in such a nervous and fidgety fashion that Mrs. Horntail declared she must have St. Vitus's dance.

"I haven't any such thing," answered Mrs. Muddauber, angrily. "I haven't any time to dance. I'm nervous, that's all. Anybody would be nervous with all the work I have to do, and my mate such a lazy fellow that he never thinks of lending me a helping mandible in making my home. He says he doesn't live very long, and wants to enjoy himself while he can. Speaking of houses, I don't approve of paper ones. I always make mine of mud. I'm a mason, you see. I get one room finished, and lay an egg in it. Then I go to market to get my baby's dinner."

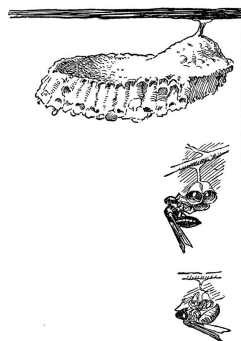
“But you haven’t any baby,” objected Mrs. Horntail. “Your egg doesn’t hatch as soon as it is laid, I know that.”

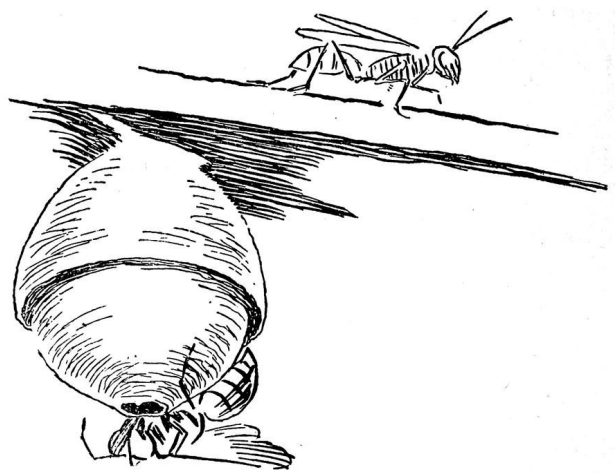
“What of it? The egg will be a baby sometime, and the baby will be hungry. He will not be a vegetarian either. He will want meat. Juicy spiders are what he prefers, and he likes them fresh. Now if I should kill them they would be anything but fresh when he is ready to eat them, so I merely sting them until they are quite paralyzed, then I put them in the room with my egg and seal it up. I build a number of cells with an egg and spiders in each, but I am not a jug builder. I have no time to fool after such silly affairs as you sometimes see on twigs and bushes.”

“She isn’t artistic enough, she had better say,” remarked the little jug builder. “My nests are wonderfully pretty. I have heard many people say so. I am very careful to give them a delicate shape. I line them with silk too, but I will not tell you how I make this silk. Even the wise men have not discovered our secret.”

“Disagreeable creature!” remarked Mrs. Horntail; “but then what can you expect from a wasp of any kind? Now who *is* making that dreadful noise? I shall certainly be a wreck before I get away from this place. People who buzz in such a fashion ought certainly to be turned out. But there, what’s the use of asking? I might know it could only be——”

“Sir Bumble Bee at your service.” And a big fellow dressed all in black and gold buzzed up before the angry Mrs. Horntail.





CHAPTER XII

THE HONEY MAKERS

Gaily we fly, my fellows and I,
Seeking the honey our hives to supply.

“I am an American,” he went on, in a voice which all could hear. “A native of this great and glorious country, and I have a right to buzz, or make any noise I please. Those little bees who make honeycomb are foreigners—immigrants. Useful citizens, I will grant, but still immigrants. Now, *my* ancestors were here when Columbus discovered America. Do you know that my name is Bombus, spelt with a big ‘B’? Now, to show you how useful we bumble bees are, I shall tell you a story. Once upon a time—are you all listening?”

“I am,” answered Ruth, quickly. “Please go on.”

“Well, once upon a time there was no red clover in Australia, and the farmers of that country decided to take American seed there and plant it. The first year the crop grew finely. There were plenty of flowers, but no seeds. Of course that was bad, they needed seed for the next year’s sowing. Well, once more they brought seed from America, and once more the crop grew finely, but not a seed came from it. Then the people began to think, and after a while they found out the trouble. They hadn’t the American bumble bee and they had to have him, for, my friends, we, only, of all the bees, can fertilize the red clover blossom, for only we have tongues long enough to reach its nectar cups and the cell where its precious pollen is hidden. You may not think our tongue so long, because it is rolled up when we are not using it, but look!” And he unrolled a long brown tongue, which, in a moment, seemed gone again.

“Gracious!” said Ruth.

“Now do you wonder that we can reach down into the red clover? When we went to Australia the clover not only grew, but set seeds too.”

“But,” questioned Ruth, “do different flowers have different bees to come to them, and how do you know?”

“Ah, that’s just it. A voice within us seems to whisper, ‘Go to the blossom whose heart you can best reach, feed upon its honey and take your fill of its golden dust.’ We know it to be the law, and we obey, and, even as we obey, the pollen clings to our hairy bodies, and we bear it to the next flower we visit. This is what usually happens, but sometimes,” he added, as though ashamed, “I must say, we break the law, and, finding a flower whose honey we cannot reach, we use our tongues to cut a hole in the spot where we know the nectar is hidden and enter from the outside. Plainly speaking, it is the way of the thief, getting our feast without paying for it. For the bee who takes it so carries away no pollen, and an honest bee should never act so. Now perhaps you would like to know how we bumble bees began life? I am sure the little girl would.” And Ruth nodded an emphatic “Yes.”

“We do not live all Winter, as honey bees do. Only a few queens sleep through the cold months, and they do not need food; so while we make a little honey to eat in Summer, we do not lay by any stores for Winter, and naturally we make no combs. What looks like them are the silken cocoons our babies spin. If I were a queen, I wouldn’t be here. Queens have too much work to do to be abroad in Summer. You may see them in the early Spring flying about and hunting up good home sites. A hole under a log is often chosen, and gathering nectar and pollen the queen carries it to this underground palace. In the mass she lays an egg, then gathers more, in which she also lays an egg. In this way her house is soon full. When the eggs hatch, the babies eat the pollen and nectar they find around them. I was just such a baby, and, being a gentleman, I haven’t much to do. I shall probably marry a queen some day, but now I simply play in the sunshine. We bumble bees belong to the social branch of the family, but there are many bees who live alone. They all follow trades. There is the carpenter, who isn’t furry like us, but black and shiny. She can bore right into solid wood and make cells for her eggs. Then there are the miners, who burrow into the ground, and the masons, who make nests out of grains of sand glued together, or out of clay or mud. Some of the carpenters line their nests

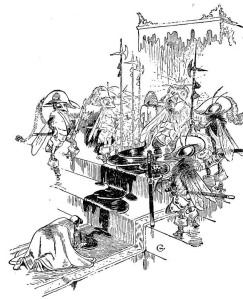
with pieces of leaves, which they cut out with their sharp jaws. They have been called upholsterers and they——”

“This is all very interesting,” interrupted a honey bee, “but really I must speak now. I have so much to say, and my work is waiting.”

“Talk, by all means,” answered Sir Bumble Bee, gallantly. “I am a gentleman, and I always yield to ladies.”

“Thank you, but I can’t call myself a lady. I am just a worker honey bee. My name is Apis Mellifica, but I do belong to a wonderful family. I will admit that. We are the greatest wax makers in the world. I heard somebody once say that bees are always in a hurry, while butterflies seem to take their time. Now there’s a good reason for that. Butterflies haven’t any work to do. They do not even see their children, and never take care of them, while bees have thousands of babies to feed and look after. Then you must know we clean house every day, for we are extremely neat housekeepers. We clean ourselves also, and we have combs and brushes for that purpose.”

The words combs and brushes seemed to have quite an effect on the bees and ants in the audience, and many began to make their toilets, Miss Apis among them. They looked so very funny that Ruth laughed outright, but she quickly settled down to listen, as Miss Apis, feeling herself quite clean, said briskly:



THE QUEEN BEE
AND HER
BODYGUARD OF
DRONES

“Now I will tell a story. Once upon a time there was a large hive under an apple tree. A hedge sheltered it from the wind, and the tree shaded it from the sun, which made it very pleasant for the family who lived there. It was a very large family, for there were thousands and thousands of members, but

they lived together in peace, each doing her own share of work. Of course there was a queen. She had a long, slender body and short wings. This did not matter, for she had only flown from the hive once, and then she had a bodyguard of drones. Maybe you think that because she was a queen she had nothing to do. It is true, she was not obliged to gather honey, make wax, clean house, nurse the children, or anything of that sort, but she was kept busy laying eggs. She laid thousands every day.”

Ruth opened her eyes wide. “Think of it, Belinda!” she said. “Thousands of eggs a day! Just suppose she was a hen.”

“She is something far more important,” answered Miss Apis, “and her eggs are of much more consequence. Besides the queen there were drones and workers in this big family. The drones did no work at all, though they were large and thick-bodied. Indeed, all they seemed fit for was to fly with the queen when she took her one trip abroad, and to eat what the workers gathered.”

“See here!” said a drone from the back of the assembly. “I am getting tired of being called lazy. I should like to say right here that we drones haven’t any honey sac nor any pollen baskets, not even a pollen brush, like Mrs. Carpenter Bee, so how can we gather pollen or honey? Besides, we haven’t any sting to defend ourselves with.”

“We will not argue the point,” said Miss Apis, “but go on to the workers, who formed the largest part of the colony. They were hatched to work, and they were willing to work until they died. They had strong wings, lots of eyes, and three stomach sacs.”

“Well, I can’t see any use in so many stomachs,” said Mrs. Horntail, and Ruth agreed with her, though she did not say so.

“You would if you were a bee,” said Miss Apis, mildly. “You see, or maybe you don’t, that eating honey, and just swallowing it, are two different things. When a bee just swallows honey it passes through the strainer, or fine hairs, in the first sac, so that every speck of pollen may be taken out, and into the second one, where it remains until the bee is ready to unswallow it in the hive. But when a bee wishes to eat this honey it passes on into the third sac, or the real stomach, and is digested.”

“Well, I am sorry I spoke,” said Mrs. Horntail, “for I certainly do not enjoy these details.”

“I can’t help that,” answered Miss Apis, undisturbed, “I am telling facts. Not only had these workers three stomach sacs, but they also had pollen baskets on their hind legs, for it is from the pollen gathered in the flowers and mixed with honey and water that the bee bread fed to the baby bees is made. Not all the workers gathered honey, though. Some made wax and built combs, and this was a very hard job, for they were obliged to hang from the ceiling and pick wax from the under side of their bodies, then chew it and plaster it to the walls. This wax is in eight scales, or pockets, on the under side of the worker bee’s body, and it is made by what she eats. When the pockets of one bee were emptied, the next one took her place, and when the lump on the side of the wall was large enough another set of bees formed it into cells. Of course you know that the cells in a beehive are always six-sided. That is because six-sided cells use all the space, and are also strongest. At least the wise men say that is probably the reason why we make them so, and they think they know. Other of the workers took care of the babies. They fed them and kept them clean, and some aired the hive.”

Ruth’s eyes were big with questions. Miss Apis saw and continued:

“They did this by moving their wings rapidly as if they were flying, and when many did it at the same time the good air was driven around the hive and the bad air out. Then, of course, there had to be sentinels to speak to every bee who passed in, and make sure she had the right to enter, for human people are not our only robbers. There are flies that look much like us, but ask them to show their pollen baskets, and they can’t do it. Now it happened one Spring in the hive I am telling you about that the queen heard a sound that she didn’t like at all. It was a thin piping, and it came from one of the brood cells, which is the nursery of the hive.”

“‘It sounds like a young queen,’ she said, ‘but I have laid no queen eggs.’ The workers stopped their tasks long enough to talk about it. They knew perfectly well that it was a young queen, and they also knew how she happened to be there, even though the old queen had laid no eggs in the cells on the edge of the comb meant for queen eggs. The old queen did not wish another royal lady, but the workers knew that if anything happened to the old queen there would be none to take her place, and such a thing must not be allowed. So they had taken down two waxen walls between three small brood cells, where a worker egg lay, and so made it into a royal cell. They bit away the wax with their jaws, and pressed the rough edges into

shape with their feet, and when the egg within hatched, instead of feeding the baby with flower dust and honey and water, as they would have done had they intended it to grow into a worker, they fed it royal jelly. And so after it had grown and spun a cocoon, within which it had lain for sixteen days, it had become a young queen, ready to leave her cell. But the workers knew it would never do for her to come out just yet, for she and the old queen would have to fight, and one would surely die.”

“Oh, how dreadful!” cried Ruth. “Why should they?”

“Because only one queen may reign in a hive.”

“‘We will keep her in her cell a little longer,’ the workers said to each other. And they built a wall of wax over her door, leaving only a hole large enough for her to thrust out her tongue so that they might feed her. But though she couldn’t get out, she could complain.”

“I should have complained too,” said Ruth.

“Well this young queen complained in earnest, and the old queen heard her, and of course she tried to get to the cell of this pert young one, and settle her for all time. This the workers would not allow. They would not touch their old queen, but they formed a bodyguard about the cell of the new one, and so protected her.”

“‘Well,’ said the old queen at last, ‘I can’t stand this. I will not stay here. I shall take my friends with me and fly away to a place where only I shall be queen.’”

“She grew more and more excited, as time passed, and, as many of the workers were excited too, the hive was in much confusion.”

“‘We are much too crowded,’ said some of the workers.”

“‘I can’t seem to settle down to work,’ answered others. ‘What can you expect when thousands of children are added to a family in a week? The time comes when the house must be made larger, or some of the members must move.’”

“‘We will *move*,’ said the old queen in a tone of decision. ‘We will move right now. Those who are my friends, come. The others may stay with the piping thing in yonder cell.’”

“And without further words, the old queen flew away, followed by a great many workers.”

“Now I know what swarming means!” cried Ruth. “I used to wonder about it.”

Miss Apis nodded.

“When the swarm was well away, the workers who were left in the hive hastened to let out the new queen.”

“She must have been glad,” said Ruth.

“Very likely,” agreed Miss Apis. “She began her reign with a flying trip into the world with the drones. But after this, she came back to the hive, and settled down to the business of egg-laying. Of course the workers took up the same old tasks, for whatever happens, workers will work. That is why they have no love for the drones, and when Winter comes they drive these lazy ones from the hive.”

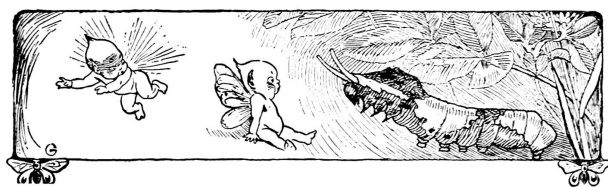
“I think I feel a little bit sorry for the drones,” said Ruth, “if they can’t help being lazy, as that drone said a while ago.”

“Well, it is our way,” answered Miss Apis. “Only those who have worked in the Summer have a right to eat in the Winter. Now my work is calling me, and I must leave. This story of one hive is true of all. I hope you have enjoyed it, and so good-by.”

“There, she is finished at last,” said Mrs. Horntail. “I think this whole meeting has been most tiresome.”

But Ruth did not agree with her.





CHAPTER XIII

THE MOST BEAUTIFUL OF ALL

Lo! the bright train their radiant wings unfurl.

—*Anna L. Barbould.*

“It seems nothing but butterflies!” cried Ruth, running out into the garden as soon as breakfast was over.

“Of course,” answered a voice, “the Lepidoptera will meet by the summer-house.”

“Does that mean butterflies? And oh, please, may I come?”

“Yes, to both questions,” was wafted back from the beautiful creature flitting so gracefully on the light warm breeze.

“Just like a flower with wings,” thought Ruth as, holding Belinda closely, she followed as fast as she could go.

Indeed, they all seemed like flowers with wings, she decided, as she came into the middle of the gathering.

“It is the most beautiful we have been to yet,” she whispered to Belinda, “and I am sure it is going to be the most interesting. I couldn’t begin to count them.”

Ruth might well say this, for nearly all the fifty-four families of moths to be found in America north of Mexico were represented by at least one member, while there were many from the four families of butterflies and the two families of skippers.

Ruth came only just in time, for already one of the moths had begun to speak. He was a handsome fellow, with fore wings in different shades of olive.

“My friends,” he said, “I am called the modest sphinx, and, that being the case, you may imagine how painful it is for me to put myself forward in this way. I have been asked, however, to give you a few general facts. Why I am expected to know these facts is, perhaps, because, being a sphinx, I should also be wise. Yet I am not the only sphinx here, and, if I remember aright, the old and historic sphinx *asked*, rather than *answered*, questions.”

“He uses awfully big words,” Ruth whispered to her usual confidant, Belinda.

“Now to begin,” went on the sphinx, “you know, I suppose, that we belong to the order Lepidoptera, which means the scale wings, because the colour of our wings is made by scales so tiny that they are really like dust. We are divided into moths, butterflies, and skippers, and all of us are messengers for the flowers, carrying the precious pollen from blossom to blossom. Our children are generally enemies to the plants. They are called caterpillars, and seem to have a great many legs, but really only six of them are true legs and remain when the youngster is full grown. The others are prolegs. There may be two or there may be ten. They help in walking, but are shed with the last skin.”

“Alas!” sighed a voice in the corner. “I haven’t any to shed—that is, in the middle of my body.”

Ruth turned as Mr. Looper, otherwise known as the measuring worm, made this remark. She would have asked a question, for Mr. Looper, rearing his head after his own queer fashion, seemed quite ready to talk, but the sphinx stopped her.

“This is not the time to talk about individual legs,” he said. “We are trying to get at general differences. Now there are certain ways in which all moths differ from all butterflies.”

“I should say so,” said Miss Papilio, a handsome tiger swallowtail. “Moths have short, stout bodies, and ours are slender.” And Miss Papilio circled above them so that all might admire her delicate body and the beauty of her tawny yellow wings, with their gray bands and stripes, and their ends pointed in true swallowtail fashion.

“And here is another difference,” she added, coming to rest with her wings folded together vertically. “We always carry our wings so when we are not flying. You moths hold yours horizontally, or sloping. Never upward.”

“Well, that’s true,” said the sphinx, “and you know we generally have beautiful feathery antennæ, though I, and a few others, are an exception to that rule, but you butterflies can boast only very thread-like antennæ, with a knob at the end.”

“Enough about that subject,” spoke up Miss Papilio. “What I am wondering about is why moths like to fly at night, or in the twilight. Now, butterflies must have sunshine.”

“We love the cool, soft night, I can’t tell you why,” answered the sphinx, “and we sleep through the noisy day.”

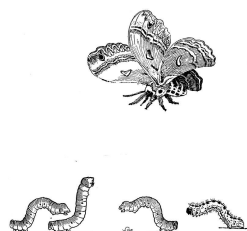
“But it is so dangerous to sleep as you do, when birds and other nuisances are up and doing.”

“Well, birds are pests, there is no doubt about it, and if it hadn’t been for them we insects would have possessed the earth long ago, but you forget, we always choose a place that is nearly the colour of ourselves, and we look so much like our surroundings that it would take a sharp eye to find us. We are not brightly coloured, as a rule, like the butterflies, or if we wear gay colours at all it is usually on our hind wings, which we hide under the fore wings. Now the general remarks being made, the audience may view the exhibits and hear their individual histories.”

Ruth was up in a second.

“I must talk to that funny measuring worm,” she said to herself. “Why, where is he?” she added, standing before the bush on which she had seen him a while before.

“Right here,” answered what Ruth thought was a twig, and which proved to be none other than Mr. Looper himself, who raised his head and began to walk on his hind legs in his own eccentric fashion. Indeed, not only he, but a number of other Mr. Loopers, all showing themselves in different positions.



“‘SMART
CHILDREN,
AREN’T THEY?’
ASKED SOME
MOTHS”

“Smart children, aren’t they?” asked some moths, variously coloured in black and brown and yellow, hovering above the tree where the loopers were feeding. “They are ours—that is, not exactly ours, but ours will be like them when they are hatched. These fellows will soon make little cradles of leaves and go into the ground to go to sleep, and when they come out they will be like us. Wonderful, isn’t it?”

“Yes,” agreed Ruth, “but I’d like to know about their legs.”

“I can explain that,” said Mr. Looper quickly. “I have no legs in the middle of my body, and as that part of me isn’t supported, I can’t walk like other caterpillars, for I *am* a caterpillar, even if they *do* call me a worm.”

“The legs, or the want of them, is a fault of his ancestors no doubt,” interrupted a voice. “Probably they walked in his idiotic fashion for fun, or to be different, even when they did have the right number of legs, and so lost the use of them, and the legs, too, finally. That often happens. I could tell you of cases——”

“Why, you look something like Miss Papilio,” said Ruth, turning to the last speaker, and interrupting her reminiscences.

“I am a Miss Papilio,” was the answer, “but not the one you heard a while ago. She was a tiger swallowtail, while I am a black swallowtail, different, but quite as handsome in my way. We swallowtails all believe in dressing well. We are butterflies, not moths, but though I am so beautiful, I serve some very humble plants. I carry the precious pollen for them. My children, I’m afraid, will not be so helpful, but what can one do? I happen to like honey, but they prefer the leaves of parsley, carrot, celery, and such things. They have large appetites, too.”

“Everything seems to have an appetite,” said Ruth.

“Well, my children will be able to eat, I can tell you. See, I have laid my eggs on this bed of parsley. Ah! there’s a larva now. Not mine, but mine will be like it. See, he is green, ringed with black and yellow. If you tease him he will stick out his yellow horns at you, and you won’t like the odour either. Would you believe I was once like that, and I slept in a pupa case like the one under the twig there? You know there always comes a time in the life of every caterpillar, if he lives long enough of course, when he stops eating for good and wants nothing so much as to sleep. That came to me, and I crawled from the parsley bed to an old rail fence and began to spin. The silk was in my body, and it came through two tubes in my lower lip.”

“That isn’t the way spiders spin,” said Ruth. “They——”

“I was not a spider,” said Miss Papilio. “I was a caterpillar, and they always spin with their mouths. So that is what I did, and before long I had lashed myself securely to the fence by strong silken loops. Then I shed my pretty suit, and my skin shrivelled until it was a hard case. In that safe cradle I went to sleep, and came out in the Spring with six legs instead of sixteen, a slender tongue in place of sharp, hungry jaws, and, best of all, four beautiful wings. Oh, the joy of sailing through wonderful space, and sipping nectar from the sweetest flowers!”

“We have all felt that way,” said a large red-brown butterfly, whose wings, lighter below, were veined and bordered by black, with a double row of white spots on the edges. “Look at the chrysalis from which I came, and say no more. Can you guess my name?”

Ruth was obliged to confess that she could not.

“I have often seen you though,” she added, “or butterflies just like you.”

“Probably you have. I am called the monarch, and, frail as I look, I can fly hundreds of miles without resting. I was just laying some eggs on this milkweed, and since you are here, you might use your eyes a little. You may see something worth while.”

Ruth was using her eyes as best she could, and soon she spied a number of caterpillars chewing away upon the milkweed leaves. They were lemon or greenish-yellow, banded with black.

“Will they grow into butterflies like you?” she asked.

“Yes,” was the answer, “but there is something more to see.”

Again Ruth looked, and now saw what appeared to be a little green jewel dotted with golden nails.

“Oh!” she cried, “how lovely!”

“I thought you would say that,” and the monarch fluttered her wings proudly. “That is our chrysalis, the cradle in which we sleep for our great transformation. That is one thing the viceroy can’t do, though she mimics us as much as possible.”

“Mimics you?” repeated Ruth, in surprise.

“Yes, certainly. You see we monarchs are wrapped in a magic perfume—that no birds like, and so they never try to eat us. Now, Mrs. Viceroy hasn’t this perfume, and to protect herself she tries to imitate our family colours, so that the birds, mistaking her for one of us, may leave her alone too. She even flies as we do. See her over there? She is smaller than I am, but quite like me, except for the black line on her hind wings. A careless observer would scarcely notice that, however.”

The monarch floated off to lay some more eggs, and Ruth found herself in the midst of ever so many tawny brown butterflies, all bordered and checkered with black, and having wings covered with silver spots.

“Oh, you are so lovely!” she cried, with shining eyes, and then, as they passed on, calling back their name, “Fritillaries!” “Fritillaries!” she turned to see many other dazzling creatures fluttering about her. Some she had never seen before, but others were like old friends. There were the meadow browns, the stout-bodied coppers, the slender, beautiful blues, and more white cabbage butterflies than she could count. The handsome red admiral flirted with the pretty painted lady, and the mourning cloaks, with their purple-brown wings, yellow-bordered and marked with light blue spots, were flitting about, telling everybody how they had slept all Winter as butterflies, which is most uncommon in the butterfly world, and were for that reason the first to show themselves in the Spring.

“I used to wonder why you were out so early,” said Ruth, “and once I found one of you in a crevice on a Winter day, and I couldn’t understand about it.”

“Well, you do now. We hibernate like many animals.”

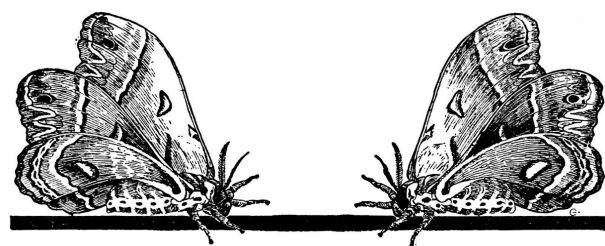
“But you must have been eggs in the beginning,” said Ruth. “The oil beetle told me that all insects begin as eggs. And will you please tell me

how a butterfly knows the right kind of plant to lay her eggs on? It always seems to be just the one her caterpillars like to eat. She doesn't eat it herself."

"Of course not," answered one of the mourning cloaks. "You need but look at our tongues to see that we eat only honey. I can't answer your question, for none of us knows. Something tells us the proper plant for our eggs. We lay them there without hesitation, and we lay a great many. This is necessary, for one never knows what may happen. Most of them may make a meal for something before they even hatch into caterpillars, and if some miss this fate, and do hatch, there are any number of birds, and their enemies, who like nothing so well as a fat, juicy caterpillar for dinner. Then if that danger is escaped, there are the birds again, and other hungry things, all anxious to get a taste of the butterfly. So you can understand that in a life so full of accidents it is important to have many eggs to begin with."

"Yes," said Ruth, "but——"

She didn't finish, for just then she put her hand on what she thought was a leaf, and, much to her surprise, she found that it was alive.



CHAPTER XIV

REAL FAIRIES

or the possible glory that underlies
The passing phase of the meanest things.

Mrs. Whitney.

Alive it certainly was, this exquisite green moth, which rose on shimmering wings at Ruth's touch. No wonder Ruth almost screamed aloud in her surprised delight.

"Are you a moonbeam?" she asked. "You are just lovely enough for one."

"No, I am not a moonbeam," was the answer, "*but I am the moon moth, the Luna.* I am a messenger for the night-blooming flowers, for only the long tongues of the moths may reach through the deep tubes to their honeyed hearts. I was taking my day nap when you touched me."



“I AM THE
MOON MOTH,
THE LUNA”

“I didn’t know you were there,” said Ruth, “you looked so much like a leaf.”

“That is what I wished to look like. Many others are sleeping the same way. You wouldn’t know them unless they moved. Our larvæ are not sleeping, however. I can answer for that. They are quite awake and busy eating the leaves of hickory, walnut, and other trees of that family. Maybe you have seen them? They are large and handsome, and they spin very snug cocoons of silk, wrapped about with a dead leaf, very much like those made by the polyphemus babies.”

“Now you know your cocoon never had the quantity of silk in it that mine had,” said a yellowish-brown moth, rising from the trunk of a nearby tree.

She was very handsome. There were window-like spots on her wings, and dusky bands edged with pink. Not far away were her larvæ, having a good time chewing the leaves of a plumb tree. They were light green, with an oblique yellow line on each side, and a purplish-brown V-shaped mark near the end of their bodies.

“You may always know the polyphemus children by that mark,” said Mrs. Polyphemus, for it was she who had interrupted the Luna’s remarks. “Now, speaking of cocoons,” she went on, “as I said before, ours contain a

great deal of silk. They have been used in the making of silk too. Shall I tell you my story?"

Of course Ruth wanted to hear it.

"Very well," said Mrs. Polyphemus. "I belong to the family of giant silkworms, though, of course, we are not worms. I began my life on an elm leaf. It was a lovely morning in May when I was hatched, and the world seemed a beautiful place to live in. I did not spend much time admiring the scenery, though, for I was hungry. I ate the shell of my egg for the first course, then I began to chew elm leaves, and I kept it up steadily. Naturally I grew, and I changed my skin five times. When I was ready to make my cocoon I found a twig on the ground among the dead leaves, and spun a fluffy mass of gray-white silk all about it, and this wrapped in a dead leaf _____"

"What?" interrupted Mrs. Cecropia, "spin your cocoon on the ground? What a careless habit. Why not fasten it to the twig of a tree or——"

"Inside a curled leaf?" added Mrs. Promethea. "That is the safest way. The wind will rock it and——,"

"I said nothing about curled leaves," answered Mrs. Cecropia. "I never use a curled leaf. I leave that for the leaf rollers. I——"

"Well, I know swinging would make me ill," declared Mrs. Polyphemus, "and I prefer the ground for my cocoon."

"Quite right," agreed Mrs. Hummingbird Moth. "The ground for me, too. Our children always go down and——"

"Gracious! you don't suppose my children would go down in the ground?" asked Mrs. Polyphemus. "No, indeed; they will sleep in their cocoons, among the fallen leaves on top. It is snug and cozy too, this cocoon, or it will be, I should rather say, for it isn't made yet. I remember mine though. A mass of coarse silk first, and a coating of varnish inside, then more silk, and another coating of varnish. I slept soundly, I can tell you, and when I awoke in the Spring I had only to send from my body a milky fluid, which softened the varnish and silk, until a doorway was made for me to come out of. I felt very weak, miserable, and forlorn just at first. I had but six legs, and my wings seemed of no use whatever, but after I had hung a while to a twig, and my wings had grown dry and strong, I was a different being. My body was lighter and smaller too. Do you know why?"

The question came suddenly, and Ruth, though she had been listening intently, could think of no answer.

“Because the fluids from it were pumped into my wings,” said Mrs. Polyphemus. “The next time you see a moth just out of its cocoon, hanging by its feet and waving its wings to and fro, you may know it is pumping fluids into them, so they may grow big and strong. You may see many wonderful things if you only keep your eyes open. Well, to go back to my story: After my wings were strong, I could fly and be as happy as I pleased. Now it is time for me to lay my eggs.”

“I wondered if you ever meant to stop talking,” said Mrs. Promethea. “There are others, you know. I really can’t see how you Polyphemuses grow up, considering the careless way your cocoons lie about on the ground. Perhaps the people who say that caterpillar children are not cared for have you in mind. Generally I believe it is better for children to help themselves. You never hear caterpillars say, ‘I can’t do this, and will some one please help me to change my skin, or some one spin my cocoon for me?’ No, they do these things for themselves, and ask no advice about them either. Still I do believe one can’t be too careful about cocoons, for once you are in one and asleep you can’t defend yourself. It is much better to make them safe to begin with. That was what I thought when I made mine. I enclosed it in a leaf, and then to make sure the leaf wouldn’t fall in the Winter winds, I fastened it to a branch of the tree with a thread of silk. No wind or anything else could break that thread. It was so strong. Just try it,” she added to Ruth, “the next time you find a Promethean cocoon. You will probably see a number together, but all will have the same strong fastenings. Another thing, I didn’t have to make a hole to get out by, as Mrs. Polyphemus told us she did. My cocoon had a valve in the top, and I had only to crawl through that. Talk about difference in looks! My mate is so unlike me you would think he belonged to another species. Our children are very handsome. Fully two inches long and blue-green in colour, not to mention the row of lovely black knobs along their bodies.”

“They can’t compare with ours,” said a fine cecropia, settling on a branch and spreading her beautiful wings.

She was very large and very handsome. Her wings were grayish, with many markings of white, brick-red, pink, and violet, and with splendid eye spots on each.

“We are the largest of the giant silkworms,” she said, “and our larvæ are as handsome in their way as we are in ours. You can see them on the plum trees over there. They are wearing their last suits, of course, for, like all caterpillars, they eat so much they need bigger skins every little while.”

“They *are* pretty for caterpillars,” agreed Ruth, looking at the blue-green creatures, with their knobs of red, yellow, and blue, all bearing black bristles.

“They are pretty enough for *anything*,” declared Mrs. Cecropia, with decision. “Our cocoon is large and fine too. Indeed, everything about us is first class. We never enclose our cocoon in a leaf, though sometimes a dead leaf may cling to the outside. We spin it along a branch, to which it is securely fastened. Some are larger and looser than others, but all are beauties.”

“Well, *I* can’t boast of fine clothes,” said a plainly dressed little moth, who was quietly hiding on a shrub, “but I belong to a very old family, and a very useful one. We were known and appreciated in Asia more than four thousand years ago. I, too, came from a tiny egg. My body was black, covered by stiff hairs, and of course I was hungry. I liked best the leaf of the mulberry tree, and I ate so much I had to change my dress often, as all caterpillars do. They all get too big for their skins, and that is what I did, but, finally, I lost my appetite, and I knew the time had come for me to spin my silken cradle. And now I may boast with good reason, for I am the true silkworm. My cocoon is spun in one thread a *quarter of a mile long*.”

“Indeed!” said Mrs. Cecropia. “I should like to know how you measured it.”

“I haven’t measured it,” the silkworm answered, “but the wise men have. Not my particular cocoon, you understand, but those of our family, and they are said to average that. They are very pretty too, these cocoons. I suppose you have all seen them? I was nine days making mine, and three days after that I cast off my baby clothes and went to sleep. I was very weak when I awoke and left my cocoon cradle, but I soon grew stronger and could walk, for you must know that the family to which I belong is not in the habit of flying. Its members are homebodies and seldom use their wings. Many of us, I may say the majority, do not live to be moths, for our cocoons are so precious, because of the long silk thread, that the larvæ are killed before they come out.”

“Why?” said Ruth.

“Because when the larvæ come out they break the thread. And now perhaps you understand how very useful we are, for all the silks, satins, ribbons, and velvets in the world are made by us.”

Ruth’s eyes grew wide with astonishment.

“It is a big boast, isn’t it?” said a very small straw-coloured moth, flitting rapidly about. “It is a true one, though. My children make cocoons too, and I made one myself, but it was quite unlike a silkworm’s, and I have an idea we are not considered useful either. I do not work among the flowers. I belong to the Wool Exchange, at least that is what somebody said about me once. My eggs will not be laid on a plant, or any growing thing. I shall choose carpet, or fine cloth, or something of that sort, and when my babies hatch they will gnaw away the fibres of the cloth, and eat and eat. Then what they don’t eat they will use to cover themselves with, binding the threads together with silk from their own bodies.”

“I know you, anyway,” said Ruth. “You ate my Winter dress full of holes. At least it was some moths like you.”

“No, my dear, not moths, but their caterpillar babies did the eating.”

“Well, it wasn’t nice, whoever did it,” declared Ruth, with some heat.

“Nice?” repeated Mrs. Clothes Moth. “I suppose it is nice to kill the silkworm babies and make dresses from their cradles, and nice to do a lot of other things that I could mention. I guess you had better not talk.”

Ruth was silent. She felt she had the worst of the argument.

“You must not mind,” whispered a large and beautiful moth whose wings were of many delicate shades of ash-gray marked with black.

Ruth turned to the speaker.

“You are something like the sphinx moth,” she said.

“Yes. I am a sphinx,” was the answer. “All of us look somewhat alike, though some are smaller than others, and colours vary. But our wings are always clear cut, our scales close fitting, and our colours quiet; a tailormade air about us, as it were. We are sometimes called hawk moths, because our wings are narrow, long, and strong, and sometimes hummingbird moths, because we fly at twilight, and poise above a flower while extracting its honey, just as hummingbirds do.”

“But why are you named the sphinx?” asked Ruth. “You haven’t told me that.”

“Well, you see, our larvæ have a queer habit of rearing themselves up in front and remaining in that position, and the wise men thought they looked something like the old Egyptian Sphinx. There’s a sphinx moth caterpillar on that tomato vine.”

“He is awful fat and green,” said Ruth. “Can you show me his cocoon?”

Even the larva laughed when Ruth asked this question.

“Dear, dear! what ignorance!” said the moth. “Just put your hand in that soft earth and take out what is there.”

Ruth obeyed, and presently brought up a dark brown case, pointed at each end.

“That is our pupa case,” explained the moth, “and in it is wrought our wonderful transformation. We do not weave cocoons, but the little brown case holds the same miracle of life and growth.”

“Well, there is just as much life and growth under my old blanket as in any pupa case, or cocoon, that was ever made.”

The speaker was a hairy caterpillar, chestnut brown in the middle, and black at each end.

“That’s the woolly bear,” explained the sphinx. “Just pick him up, and see what will happen.”

“I know,” answered Ruth, but nevertheless she took the little brown fellow in her hand, whereupon he promptly curled up in a tight ball and rolled to the ground.

“I will do it every time,” said the caterpillar. “I have been called the hedge hog because of that cute trick.”

“It is cute,” agreed Ruth, “but what do you mean by your blanket?”

“Oh, as to that, I don’t fool after cocoons, or pupa cases, or the rest of it. I simply take off my hair when I am ready for my long sleep, and make it into a blanket, which covers me snugly.”

“But it is a cocoon just the same,” persisted Ruth.

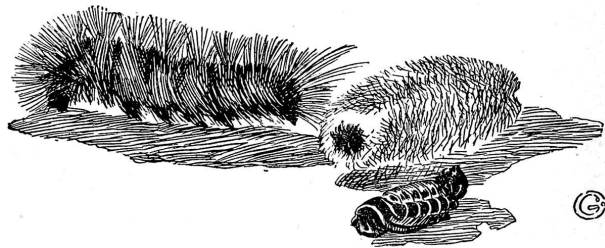
“Well, you may call it what you please, I say it is a blanket. When I wake from my sleep under it I am no longer a caterpillar, but a moth.”

“Like me,” added a dull yellow moth, spreading her black dotted wings. “I am the Isabella, if you care to know.”

“So you see,” rejoined the woolly bear, “it really doesn’t matter whether it is a cocoon, a pupa case, or a blanket which encloses the glory of our transformation, the marvel of it is just the same.”

Long after they had drifted by, that gay company of butterflies and moths, Ruth sat thinking of the wonder of it all.

“Didn’t I tell you, Belinda,” she whispered, “didn’t I tell you it was really living in Fairyland, and now, when we can hear what they say, and they tell us such interesting things, it is more Fairyland than ever. The Wind told us to watch and listen, and we will do that. We will watch and listen with all our might, for oh! Belinda, there is such a lot to learn yet.”



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